

Koebe's Circle-Domain Conjecture

OpenAI

September 23, 2026

Abstract

We prove that every domain in the Riemann sphere is conformally equivalent to a circle domain, resolving Koebe's circle-domain conjecture positively.

Contents

1	Introduction	3
1.1	From finite uniformization to boundary control	3
1.2	Proof overview	4
1.3	Conventions and classical inputs	5
2	The quotient sphere and compatible coordinates	5
2.1	Collapsing complementary components	6
2.2	Coordinates, bands, and compatibility	8
2.3	The geometry of generic level unions	10
3	Finite transfer and normalized limits	12
3.1	The transfer theorem	12
3.2	Circle-domain completions and compactness	17
3.3	Identification and trapping of the ends	19
4	Tagged paths and compatible barriers	20
4.1	Endpoint data and barriers	20
4.2	Path length, tags, and residual contacts	21
4.3	The compatible barrier theorem	23
4.4	Cutting, joining, and coding paths	23
4.5	Collar concatenation	25
4.6	Removing a newly introduced level of small jumps	27
4.7	Proof of the compatible barrier theorem	28
5	Positive laws of admissible paths	32
5.1	Duality for a Borel family of measures	33
5.2	Countable and atomless tests	35
5.3	Fusion from countable tests to the whole dust	37
5.4	Application of fusion to path laws	39
6	The first selection and path-level intersections	41
6.1	Point avoidance and intersections of two laws	41
6.2	The crossing argument	43
6.3	Selecting barriers and collapsing unmarked components	45
6.4	Free contacts with prior levels	47
7	Period coordinates and the marked components	49
7.1	Reduction to a compact family of simple arcs	50
7.2	Averaging the index of a ray	51
7.3	A boundary estimate for a transferred stream	55
7.4	Excluding a protrusion	56
7.5	Selection with prescribed retained tests	58
8	A hyperbolic convex-hull realization	61

1 Introduction

A *domain* is a nonempty connected open set. A *circle domain* in the Riemann sphere $\widehat{\mathbb{C}}$ is a domain whose complementary components are closed round disks or single points. Koebe's *Kreisnormierungsproblem*, posed in 1908 [11], asks whether every planar domain has such a conformal model.

Theorem 1.1 (Circle-domain uniformization). *Every nonempty connected open set $G \subset \widehat{\mathbb{C}}$ is conformally equivalent to a circle domain.*

This proves Koebe's conjecture in its unrestricted form. The difficulty is to control all complementary components at once. There may be uncountably many of them, with no regularity or separation. Even when finitely connected approximations have round complementary disks, a limiting complementary component can contain more than the disk obtained by following one of those disks to the limit. The proof must rule out that excess and also control components for which no disk has been followed.

Our construction chooses a countable set of complementary components to mark. Write b for the label of a source component. For a limiting conformal map f , write $K_b(f)$ for the corresponding complementary component of $f(G)$. At a marked component, the finite approximations supply a distinguished round disk $D_b \subset K_b(f)$, possibly reduced to a point. We arrange that

$$\begin{cases} K_b(f) \text{ is a singleton,} & b \text{ unmarked,} \\ K_b(f) = D_b, & b \text{ marked.} \end{cases}$$

These are the two geometric conclusions needed for the theorem. Figure 1 illustrates the second obstruction. The component correspondence and the inclusion $D_b \subset K_b(f)$ are proved in Section 3.

1.1 From finite uniformization to boundary control

Koebe proved the finite-connectivity theorem; see his 1920 treatment [12]. He and Schramm established the countably connected case [7, Theorem 0.1]. Schramm introduced transboundary extremal length and gave another proof, as well as uniformization results for domains bounded by points and K -quasicircles for a fixed K [19]. These results combine finite conformal approximation with estimates controlling which boundary components survive the limit.

Geometric conditions also yield circle-domain models beyond countable connectivity. Esmayli and Rajala proved uniformization for cospread domains, and more generally under a uniform

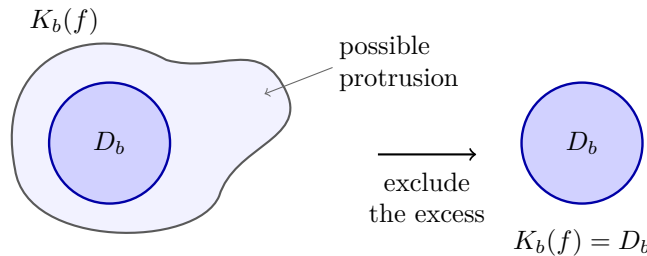


Figure 1: A distinguished disk may initially be only part of a limiting complementary component. The marked-component argument excludes every such protrusion. This is a schematic picture; no boundary regularity of $K_b(f)$ is assumed.

quasitripod condition and a packing bound [4, Theorem 1.3 and Corollary 1.6]. Karafyllia and Ntalampekos established the conjecture for spherical Gromov-hyperbolic domains, obtaining uniform circle-domain models and Möbius uniqueness [9, Theorem 1.2]. Theorem 1.1 imposes no geometric condition on the complement.

Finite approximations alone do not settle the limiting problem. Rajala showed that arbitrary interior exhaustions can fail to have circle-domain limits, and proved an exhaustion refinement theorem for countably connected domains [18, Theorems 1.1–1.3]. Combining Theorem 1.1 with Ntalampekos and Rajala’s exhaustion theorem [16, Theorem 1.1] gives an unconditional fixed-domain approximation: every proper domain $G \subsetneq \widehat{\mathbb{C}}$ has an exhaustion $G_j \subset G_{j+1}$, $\bigcup_j G_j = G$, by finitely connected domains bounded by finitely many disjoint Jordan curves lying in G . Fix three distinct points $a_1, a_2, a_3 \in G_1$. The conformal uniformizations $f_j : G_j \rightarrow D_j$ onto finitely connected circle domains, normalized by $f_j(a_k) = a_k$ for $k = 1, 2, 3$, converge locally uniformly in the spherical metric along the whole sequence to a conformal homeomorphism of G onto a circle domain. For $G = \widehat{\mathbb{C}}$, the identity gives the trivial case. This asserts the existence of a suitable exhaustion, not convergence for arbitrary exhaustions. For further history, see [15].

We obtain the required control by choosing analytic tests on G before taking finite uniformizations. A crucial freedom is to retain the original conformal structure on each compact core while choosing a new structure on its finite completion. The core eventually exhausts the whole domain, so the limiting map is conformal in the original structure. The tests are real functions or functions with additive period one around a selected component. Each has finite Dirichlet energy and a square-summable bound on its oscillation at the complementary components. Small real barriers collapse unmarked components; period-one tests detect a protrusion beyond a marked disk through a boundary integral.

The mixed use of domain energy and boundary weights belongs to the transboundary extremal-length approach introduced by Schramm [19]. Our path construction charges repeated boundary visits individually and records portions traveling in earlier level sets. The law duality is a form of Fuglede’s modulus for measure systems [6], with the probability-measure duality of Ambrosio, Di Marino, and Savaré [2]. The additional tasks here are to fuse the countable tests into one law controlling all boundary components, to preserve compatibility as tests are appended, and to exclude all possible noncircular limiting components.

1.2 Proof overview

Collapse each complementary component of G to one point. The quotient X is a topological sphere, and $B = X \setminus G$ is compact and totally disconnected. Thus B supplies the labels b used above. We use the topology of X to approach components, while retaining the original conformal structure on G for length and energy.

Section 2 defines the tests and their compatibility condition. Compatibility controls how the level sets of a new test meet continua formed from earlier levels. It permits the simultaneous transfer in Section 3: finitely many tests extend to a finite conformal completion, agree with the originals on a prescribed compact core, and acquire arbitrarily little additional energy or excess boundary oscillation. Their periods are preserved exactly. The conformal structure may change outside the core; the cores eventually exhaust G . Finite circle-domain uniformization then produces the limits and component correspondence described above.

The tests themselves come from a boundary alternative. Either an end can be separated from a fixed interior anchor by a small compatible barrier, or there is a probability law of paths from the anchor to that end with controlled mean arclength occupation. Section 4 constructs the barriers

using paths that may run through complementary components. Tags record portions lying in previously chosen level sets; the remaining boundary contacts obey explicit incidence bounds. These conditions allow paths to be spliced near their ends without losing the estimates.

Section 5 proves the alternative. We use the modulus–probability duality of Ambrosio, Di Marino, and Savaré [2], proving the required measure-family form in full. This first gives a path law for each prescribed countable set of boundary tests and each atomless test measure. A separate fusion argument produces *one* law whose hit probabilities have bounded counting $\ell^2(B)$ norm on the entire, possibly uncountable, set B . Expected visit counts on previously prescribed countable sets are controlled separately.

Section 6 removes diffuse intersections between independent path laws. A four-path crossing argument then shows that only countably many ends can lack small barriers. Selecting countably many barriers gives a countable marked set $S \subset B$ and collapses every $K_b(f)$ with $b \notin S$. This conclusion survives every later countable compatible extension of the tests.

Section 7 handles the marked components. Averaging the integer indices of lifted simple arcs produces period-one tests with small energy and small oscillation away from the selected end. A Stokes estimate turns these bounds into an upper bound on a boundary integral. A protrusion would force a strictly larger lower bound. Each such test excludes a neighborhood of a possible pair (f, D_b) ; second countability permits a countable selection excluding all pairs with $D_b \neq K_b(f)$. The final simultaneous transfer proves Theorem 1.1.

Theorem 7.5 records the same construction with an arbitrary prescribed countable compatible list retained from the outset. For a finite family of bounded continuous real functions on X , smooth with finite Dirichlet energy on G , the simple sufficient condition $\text{area}((u_i, u_j)(B)) = 0$ for distinct tests gives exact agreement on exhausting cores, vanishing tail energy, and vanishing square-summed boundary oscillations; see Corollary 7.6. This quantitative conclusion supplies the analytic input for the removable-boundary rigidity companion [17].

Section 8 derives the geometric consequence from Luo and Wu’s Theorem 1.1(a) [13]: every complete hyperbolic surface of genus zero is intrinsically realized by the boundary of a hyperbolic convex hull whose ideal set has round-disk or point components.

1.3 Conventions and classical inputs

The classical inputs are finite circle-domain uniformization, compact genus-zero uniformization, Moore’s decomposition theorem, area and coarea, and standard Borel selection and refinement theorems. They are cited where used. All constructions of compatible tests, fusion of laws, and component exclusions are proved below.

Path laws are Borel probability measures on the stated code spaces. We use their completions for universally measurable selections and analytic events, and take the resulting laws on the Borel sigma-algebra. Every nonnegative sum over an arbitrary index set is the supremum of its finite subsums. In particular, every ℓ^2 function has countable positive support.

2 The quotient sphere and compatible coordinates

We first collapse each complementary component to a point and construct smooth compact regions exhausting the domain. We then define coordinates with controlled boundary oscillation and a compatibility condition on their level sets. The last two lemmas give the level-set geometry and null-image criterion used to establish that condition.

Throughout, a domain is nonempty, open, and connected. The case $G = \widehat{\mathbb{C}}$ is immediate, so we assume that the complement is nonempty. After a Möbius change of coordinate, we assume that $\infty \in G$. Fix $R > 0$ such that

$$\{z : |z| \geq R\} \cup \{\infty\} \subset G.$$

We use ds and da for spherical length and area on the original sphere. We call this the *physical* sphere when distinguishing it from a quotient introduced below. The L^2 norm of a one-form is its Dirichlet energy norm; in two dimensions this norm is unchanged by a conformal change of metric.

2.1 Collapsing complementary components

A *continuum* is a nonempty compact connected set; it is *nondegenerate* if it has more than one point.

Lemma 2.1 (The quotient sphere). *Collapse each component of $\widehat{\mathbb{C}} \setminus G$ to a point, and leave the points of G as singleton decomposition elements. The resulting quotient map*

$$\pi : \widehat{\mathbb{C}} \longrightarrow X$$

is a closed monotone map onto a topological sphere. Its restriction to G is a homeomorphism onto an open subset of X . The set

$$B = X \setminus \pi(G)$$

is compact and totally disconnected. Moreover, the inverse image under π of every continuum in X is connected.

Proof. Put $F = \widehat{\mathbb{C}} \setminus G$. To apply Moore's decomposition theorem, we check that the decomposition relation is closed and that every element is nonseparating. Suppose first that x_n, y_n belong to the same decomposition element and converge to x, y . If infinitely many of those elements are singletons, then $x = y$. Otherwise, after discarding finitely many terms, let F_n be the component of F containing x_n, y_n . The hyperspace of nonempty compact subsets of the sphere is compact, so a subsequence of F_n converges in Hausdorff distance to a nonempty compact set $A \subset F$. This limit is connected and contains x, y . Thus x, y lie in one component of F , which proves closedness of the relation.

Now let Q be a component of F . The connected set G lies in one component of $\widehat{\mathbb{C}} \setminus Q$. Any other component V would be disjoint from G , and its nonempty boundary would lie in Q . The connected set $Q \cup \overline{V} \subset F$ would then strictly contain Q , a contradiction. Hence Q is nonseparating. Moore's theorem now identifies X with a topological sphere; see [14]. The quotient is closed and has connected fibers. Since G is open and saturated, $\pi|_G$ is a homeomorphism onto an open subset.

It remains to prove the inverse-image assertion and total disconnectedness of B . Let $A \subset X$ be a continuum. If $\pi^{-1}(A) = P \cup Q$ were a separation into disjoint nonempty compact sets, every fiber over A would lie entirely in one of P, Q . Consequently $\pi(P)$ and $\pi(Q)$ would be disjoint nonempty compact sets separating A , a contradiction. A continuum contained in B therefore has connected inverse image in F , which must lie in a single component of F . The continuum is consequently a singleton. Finally, B is the compact image of F . $\square$

We identify $\pi(G)$ with G , orient X consistently with the physical orientation on G , and fix a compatible metric d_X on X . Since a nonempty open subset of a sphere contains a nondegenerate continuum, B has empty interior. Thus G is dense in X . We refer to B as the *dust*.

A *dust disk* is a Jordan domain $U \subset X$ whose closure misses ∞ and whose boundary is a smooth physical Jordan curve in G . Its physical inverse image is exactly the bounded Jordan domain enclosed by that same curve. Indeed, the decomposition elements avoid the curve and cannot cross it, so the two Jordan sides are saturated. We shall use the same notation for a dust disk and its physical inverse image when no confusion can result.

The next lemma supplies small disks around dust points and smooth contours avoiding prescribed closed sets. These constructions will give the compact exhaustion of G .

Lemma 2.2 (Small disks and contours). *The following assertions hold.*

- (i) *If $F \subset X$ is closed, every nondegenerate continuum contained in $F \cup B$ is contained in F .*
- (ii) *Suppose that $F \subset X$ is closed and contains B , and that $\{x\}$ is a component of F , with $x \neq \infty$. Every neighborhood of x contains the closure of a dust disk containing x whose boundary misses F .*
- (iii) *Let A be a compact relatively clopen subset of B , and let $W \subset X \setminus \{\infty\}$ be open with $A \subset W$. For every $\varepsilon > 0$, there are finitely many dust disks with pairwise disjoint closures, all contained with their closures in W , of d_X -diameter less than ε , whose union contains A . The disks can be required to avoid $B \setminus A$ and to meet A .*
- (iv) *Suppose that finitely many dust disks have closures in a Jordan domain $V \subset X \setminus \{\infty\}$, and their boundaries avoid a closed set $F \supset B$. Their closures can be covered by finitely many Jordan disks with pairwise disjoint closures in V and smooth boundaries in $G \setminus F$.*

Proof. We use two facts about compact spaces. First, a component of a compact Hausdorff space is the intersection of its relatively clopen neighborhoods. Second, if C is a continuum and P is a proper closed subset of C , every component of P meets the boundary of P relative to C . Indeed, a component missing that compact boundary could be separated from it by a relatively clopen subset of P . That subset would be both open and closed in C , a contradiction. The second fact is the boundary-bumping principle.

For (i), suppose a nondegenerate continuum $C \subset F \cup B$ contains $x \notin F$. Choose $y \in C \setminus \{x\}$ and a small closed metric neighborhood N of x missing both F and y . The component of $C \cap N$ containing x meets the boundary of N by boundary bumping. It is therefore a nondegenerate subcontinuum contained in B , contradicting Lemma 2.1.

For the remaining parts, we first describe a common contour construction on the physical sphere. If A_0 is a compact subset of a physical open set O away from infinity, a smooth cutoff supported in O and equal to one near A_0 has a regular level between zero and one. This level is a compact smooth one-manifold, so it has finitely many components, each a smooth Jordan curve. The corresponding superlevel set contains A_0 and has compact closure in O . Taking its outer boundary curves fills its holes. If $O \subset \pi^{-1}(V)$ for a Jordan domain $V \subset X \setminus \{\infty\}$, each filled outer contour still lies in $\pi^{-1}(V)$. Indeed, $\pi^{-1}(X \setminus V)$ is connected by Lemma 2.1, contains infinity, and misses the contour; it must therefore lie on its unbounded side.

For (ii), take a Jordan neighborhood V of x whose closure is inside the prescribed neighborhood and misses infinity. By the first compact-space fact above, there is a compact relatively clopen set $A_0 \subset F$ with $x \in A_0 \subset V$. Choose a physical open neighborhood of $\pi^{-1}(A_0)$ compactly contained in

$$\pi^{-1}(V) \setminus \pi^{-1}(F \setminus A_0).$$

Apply the contour construction there. Its boundary curves miss $\pi^{-1}(F)$. The connected fiber $\pi^{-1}(x)$ lies inside one of the resulting outer contours. Its filled disk projects to the required dust

disk. Its closure is contained in V by the connected-complement observation above.

For (iii), the compact metric totally disconnected space B has a clopen base. Partition A into finitely many compact relatively clopen pieces $A_1, \dots, A_m$, each contained in a Jordan neighborhood V_i with closure in $W \setminus (B \setminus A)$ and diameter less than ε . The compact physical preimages of these pieces are pairwise disjoint. They therefore have pairwise disjoint open neighborhoods with disjoint compact closures, each lying in $\pi^{-1}(V_i)$ and avoiding $\pi^{-1}(B \setminus A_i)$. Make physical contours inside these neighborhoods. The boundary curves obtained for all pieces are mutually disjoint and miss the whole physical complement of G . Their filled disks are either nested or have disjoint closures. Discard the nested ones and any disk missing A . The remaining disks still cover A , have disjoint closures, and each stays inside its corresponding V_i .

For (iv), let Q be the union of the given closed disks. Its boundary is contained in the union of the given boundaries and hence misses F . Use the regular-level construction to smooth a sufficiently small physical outer neighborhood of $\pi^{-1}(Q)$. Take the smoothing collar around the boundary of $\pi^{-1}(Q)$ disjoint from $\pi^{-1}(F)$ and keep the neighborhood's closure in $\pi^{-1}(V)$. There are finitely many smooth boundary curves by the compact one-manifold argument above. Fill the outer contours and discard nested contours. The connected-complement observation keeps all filled disks in V , and all new boundaries remain in $G \setminus F$. $\square$

We now use part (iii) to fix the compact exhaustion. Additional closed obstacles can be avoided by choosing W disjoint from them, and each new cover can have its closures inside the preceding disks. Choose cores

$$K_j = X \setminus \bigcup_{U \in \mathcal{U}_j} U, \quad K_j \subset \text{int}_X K_{j+1}, \quad \bigcup_{j \geq 1} K_j = G,$$

where each $\mathcal{U}_j$ is a finite family of dust disks with pairwise disjoint closures, covers B , lies on the bounded side of $|z| = R$, and has mesh tending to zero. Here the mesh is the maximum of the d_X -diameters of the disks in the cover. Each K_j is a compact smooth bordered region in G containing infinity in its interior. Every compact subset of G lies in the interior of all sufficiently large cores. For $b \in B$, write

$$U_j(b) \in \mathcal{U}_j, \quad b \in U_j(b), \quad C_j(b) = \partial U_j(b).$$

Then

$$\overline{U_{j+1}(b)} \subset U_j(b), \quad \bigcap_j \overline{U_j(b)} = \{b\}.$$

Smooth intermediate curves and compact physical collars can be chosen between successive boundary curves. We sometimes call the curves $C_j(b)$ themselves collars; an actual annular neighborhood will always be specified when one is needed.

2.2 Coordinates, bands, and compatibility

We next specify the functions to be transferred from these cores. For each function, we record its region of definition, any additive period, and bounds on its energy and limiting oscillation. Compatibility will constrain how the level sets of different coordinates can meet at the dust.

Every nonnegative sum indexed by an arbitrary set is understood as the supremum of its finite subsums. In particular,

$$\|a\|_{\ell^2(B)}^2 = \sum_{s \in B} a(s)^2.$$

A nonnegative $\ell^2(B)$ function has countable positive support.

Definition 2.3 (Coordinates). A coordinate u_i comes with a closed set $M_i \subset X$, called its *mask*, and has one of the following forms.

- (i) A bounded smooth real function on G , with mask $M_i = X$.
- (ii) A coordinate with additive period one, with mask $M_i = \overline{U}$ for a dust disk U and singular point $p_i \in B \cap U$. On the cyclic universal cover of $\overline{U} \setminus \{p_i\}$, it is a smooth real function over $G \cap M_i$, smooth up to the boundary curve, satisfying

$$u_i(Tx) = u_i(x) + 1$$

for the positive deck transformation T .

In both cases the differential is single-valued and is required to satisfy

$$\|du_i\|_{2,G \cap M_i} < \infty.$$

For a nonsingular point $s \in B \cap M_i$, choose a local lift in the periodic case and define

$$I_i(s) = \left[\liminf_{\substack{x \rightarrow s \\ x \in G \cap M_i}} u_i(x), \limsup_{\substack{x \rightarrow s \\ x \in G \cap M_i}} u_i(x) \right].$$

We call $I_i(s)$ the *band* at s and require it to be a bounded interval with length at most $a_i(s)$, where $a_i \geq 0$ belongs to $\ell^2(B)$. For a coordinate with period, set $a_i(p_i) = 1$; no real band is required there. Set $a_i = 0$ off the mask. A change of local sheet translates a band by an integer.

A finite or countable well-ordered list of coordinates carries a countable flag set $\Sigma \subset B$ containing every positive support $\{a_i > 0\}$. Points of $B \setminus \Sigma$ are called *free*. Every coordinate defined at a free point has a unique trace there, interpreted modulo an integer in the periodic case.

To include dust points in the level sets, write $\mathbb{T} = \mathbb{R}/\mathbb{Z}$ for the circle of levels. For $t \in \mathbb{T}$, let $H_i(t) \subset M_i$ consist of the domain points with $u_i = t \pmod{1}$, the nonsingular dust points whose bands project to sets containing t in $\mathbb{T}$, and the singular point, if present. Thus the singular point belongs to every level. We use Haar measure of total mass one on $\mathbb{T}$ and product Haar measure on tuples. Thus

$$|\{t \in \mathbb{T} : s \in H_i(t)\}| \leq \min\{1, a_i(s)\} \leq a_i(s) \quad (s \in B). \quad (2.1)$$

Before stating compatibility, we record that each level graph

$$\{(s, t) \in X \times \mathbb{T} : s \in H_i(t)\}$$

is compact. At a nonsingular dust point, the lower band endpoint is lower semicontinuous and the upper endpoint is upper semicontinuous, as follows directly from their neighborhood-infimum and neighborhood-supremum definitions. The local boundedness furnished by a finite band also bounds all nearby bands in a common lift. These facts give closedness at nonsingular dust points. Smoothness gives closedness at domain points, and the definition includes every limiting level at a singular point. Since the mask is closed, the graph is closed in the compact space $X \times \mathbb{T}$.

Definition 2.4 (Compatibility). A well-ordered list with its flag set satisfies (P) if, for each label j , each nonempty finite list of lower labels $i_1, \dots, i_d < j$ with repetitions permitted, and almost every $(t, t_1, \dots, t_d) \in \mathbb{T}^{d+1}$, no free point of $H_j(t)$ belongs to a nondegenerate component of

$$H_{i_1}(t_1) \cup \dots \cup H_{i_d}(t_d). \quad (\text{P})$$

The null-set assertion uses completed product measure. To justify its measurability, consider the compact hyperspace of nonempty subcontinua of X . The conditions that a continuum contain a specified point, lie in a specified finite level union, and have diameter at least $1/n$ are closed conditions on the continuum, point, and level tuple. Projection over that compact hyperspace is closed. Taking the union over n shows that belonging to a nondegenerate component of the level union is a Borel relation in the point and tuple. Requiring the point to be free is Borel, and projection over the point makes the failure relation in Definition 2.4 analytic. It is therefore universally measurable. In particular, we may prove (P) by slicing in its level parameters; see [10].

Increasing Σ preserves (P), as does passage to a finite ordered sublist with the same flags. At a countable limit stage we take the union of the lists and of their flag sets. Every condition in (P) involves only finitely many coordinates, so previously established compatibility persists at such a stage.

2.3 The geometry of generic level unions

Compatibility concerns nondegenerate components of finite level unions. We show that almost every such union has only countably many of these components and that its subcontinua are locally connected. The proof uses finite physical length in G to control continua of fixed positive diameter in X .

We write $\mathcal{H}_{\text{sph}}^1$ for one-dimensional Hausdorff measure with respect to the spherical metric.

Lemma 2.5 (Generic level continua). *For each finite list $i_1, \dots, i_d$, almost every tuple $(t_1, \dots, t_d)$ gives a compact level union*

$$H = \bigcup_{k=1}^d H_{i_k}(t_k)$$

with the following properties.

- (i) $\mathcal{H}_{\text{sph}}^1(H \cap G) < \infty$.
- (ii) *For each $\varepsilon > 0$ there is a finite bound on the cardinality of any family of pairwise disjoint subcontinua of H having d_X -diameter at least ε .*
- (iii) *H has at most countably many nondegenerate components, and every subcontinuum of H is locally connected. Any two distinct points in a connected relatively open subset of such a subcontinuum can be joined by a simple arc within that subset.*

In addition, if $q \geq 0$ is Borel on G and $m \geq 0$ is in $\ell^2(B)$, then for each label i ,

$$\begin{aligned} \int_{\mathbb{T}} \left(\int_{H_i(t) \cap G} q \, d\mathcal{H}_{\text{sph}}^1 + \sum_{s \in H_i(t) \cap B} m(s) \right) dt \\ \leq \int_{G \cap M_i} q |du_i| \, da + \sum_{s \in B} m(s) a_i(s). \end{aligned} \tag{2.2}$$

The right side is finite when $q \in L^2(G, da)$.

Proof. We first prove the weighted estimate and finite length, then derive the topological conclusions from that length bound. Apply the coarea formula to the smooth circle-valued map $u_i \pmod{1}$ on its masked domain; see [5]. Mask-boundary portions contribute zero after integration in the level parameter: each point of that smooth boundary occurs at just one level. Thus the domain term in

Equation (2.2) equals its stated area integral. The dust term is bounded by Equation (2.1) and Tonelli's theorem; only the countable support of m is involved. Cauchy–Schwarz bounds the right side when $q \in L^2$. Taking $q = 1$ and $m = 0$ gives finite length for almost every level, since the spherical area is finite. Summing over the finite list proves (i).

For (ii), only compactness of H and the finite length in (i) are needed. Fix $\varepsilon > 0$ and choose a core K whose complementary dust disks have d_X -diameter less than ε . Every continuum $Q \subset H$ with $\text{diam}_X Q \geq \varepsilon$ meets K : otherwise its connectedness places it inside one of those disjoint disks. Compactness of the physical set $K \subset G$ and uniform continuity of π give $r > 0$ such that every closed spherical r -ball centered on K is contained in G and has image of d_X -diameter less than ε . Decrease r so these balls are ordinary embedded spherical disks.

Choose $p \in Q \cap K$. Since the closed spherical r -ball about p has quotient diameter less than ε , Q exits this ball. By boundary bumping, the component of the intersection containing p meets the ball boundary. Spherical distance from p , a 1-Lipschitz function, maps this subcontinuum onto $[0, r]$. Therefore

$$\mathcal{H}_{\text{sph}}^1(Q \cap G) \geq r.$$

For pairwise disjoint continua these measures add. Their number is at most $\mathcal{H}_{\text{sph}}^1(H \cap G)/r$, proving (ii).

The bound in (ii) now gives (iii). For each n there are only finitely many components of H with diameter at least $1/n$, proving countability of the nondegenerate components. For local connectedness of every subcontinuum, let $L \subset H$ be any subcontinuum, let $p \in L$, and take a sufficiently small radius $t > 0$. Let A be the component through p of $L \cap \overline{B}_X(p, t)$. If A were not a relative neighborhood of p in L , there would be points $p_n \rightarrow p$ in other components of this closed intersection. After passing to a subsequence those components are distinct: a finite union of components not containing p is compact and misses p . If the closed intersection is all of L , it has only one component, so this case is already excluded. Boundary bumping now shows that each of the distinct components meets $\{x : d_X(x, p) = t\}$. For large n it consequently has diameter at least $t/2$, in contradiction to (ii). Hence A is a connected neighborhood of p . Arbitrarily small choices of t prove that L is locally connected.

Finally, a connected open subset V of a locally connected metric continuum is arcwise connected. One can see path connectedness directly by chaining connected relatively open sets of small diameter with closures in V , and then recursively refining each link inside its predecessor. Connectedness guarantees finite chains between specified endpoints at every refinement. Choose the parameter subdivisions and image diameters to tend to zero. Completeness gives a continuous path between the endpoints whose image remains in V . The elementary loop-erasure theorem for a path in a Hausdorff space then supplies a simple arc with the same endpoints inside its image. These standard continuum facts are also recorded in [21]. $\square$

We have proved that generic level continua admit simple arcs locally. The next lemma converts bounds along such arcs into null images of traces, which will establish compatibility. The use of outer measure requires no measurability of the trace function.

Lemma 2.6 (A null image criterion for traces). *Let L be a locally connected metric continuum, let $V \subset L$ be relatively open, and let ν be a finite Borel measure on L . Suppose $F \subset V$ has ν -outer measure zero and a function $v : F \rightarrow \mathbb{R}$ satisfies*

$$|v(x) - v(y)| \leq A\nu(\alpha)$$

for some finite $A \geq 0$, every $x, y \in F$, and every simple arc $\alpha \subset V$ joining x and y . Then $v(F)$ has Lebesgue outer measure zero.

The same conclusion holds locally when F is covered by countably many such relatively open sets and v has a real branch on each. In particular, the corresponding image in $\mathbb{T}$ has Haar measure zero for a circle-valued trace with these local branches.

Proof. For $\delta > 0$, regularity of the finite Borel measure on the compact metric space L gives a relatively open set O with

$$F \subset O \subset V, \quad \nu(O) < \delta.$$

Local connectedness makes each component O_k of O relatively open. There are at most countably many components, because disjoint nonempty open sets in a second-countable space are countable. Any two points of $F \cap O_k$ can be joined by a simple arc in O_k . Consequently

$$\text{diam } v(F \cap O_k) \leq A\nu(O_k).$$

Cover these real images by intervals, allowing arbitrarily small summable excess lengths. Countable subadditivity gives

$$|v(F)|^* \leq A \sum_k \nu(O_k) = A\nu(O) \leq A\delta.$$

Letting δ tend to zero proves the first assertion. Apply it separately to the countably many local branches for the second. $\square$

3 Finite transfer and normalized limits

We now transfer finitely many compatible coordinates to a smooth surface with finitely many holes. The transfer preserves the coordinates and the original conformal structure near a prescribed compact core. Outside the core we choose the coordinates and the conformal structure together, so that the added energy and the excess over the given oscillation budgets are small. The key is to make pairwise wedge products small: a suitable metric then makes the sum of the coordinate energies small as well.

After proving this finite construction, we normalize its circle-domain maps and pass to limits on G . Nested source collars will identify all complementary components of the limit and locate the round disks associated with prescribed dust points.

3.1 The transfer theorem

The construction selects finitely many disjoint dust disks. Some are filled, with a conformal structure chosen below; the others have their interiors excised and are called *terminal disks*. Each terminal disk is indexed by a distinct dust point in its interior.

In expressions involving a physical completion, we use the same symbol M_i for the inverse image $\pi^{-1}(M_i)$ of a mask in X . Thus $Q \cap M_i$ below is a subset of the physical sphere.

Theorem 3.1 (Finite transfer). *Let $u_1, \dots, u_m$ be a finite ordered list satisfying Definition 2.3 and compatibility (P), with a common countable flag set Σ . The same assertion holds for a finite ordered sublist of a countable compatible list. Prescribe a finite set $F \subset B$ containing all singular points of the list, a compact core $K_0 \subset G$, a dust-disk cover from the fixed exhaustion, and $\varepsilon > 0$.*

There exist a smooth compact core $K \subset G$, finitely many terminal disks T_s with pairwise disjoint closures, indexed by a finite set $S \subset B$ containing F , and a smooth metric g on the physical sphere, with the following properties. Every terminal disk lies in a member of the prescribed cover,

$s \in T_s$, and K contains K_0 and all mask boundaries in its interior. The metric g agrees with the spherical metric on a neighborhood of K . Write

$$Q = \widehat{\mathbb{C}} \setminus \bigcup_{s \in S} T_s, \quad Q^\circ = \widehat{\mathbb{C}} \setminus \bigcup_{s \in S} \overline{T_s}.$$

Then $K \subset Q^\circ$, and Q , equipped with g , is a smooth bordered surface of genus zero. Each u_i has a transferred coordinate u_i^* on $Q \cap M_i$, smooth up to its boundary in local real branches, with the following properties.

- (i) $u_i^* = u_i$ on a neighborhood of $K \cap M_i$;
- (ii) The total energy outside K satisfies

$$\sum_{i=1}^m \|du_i^*\|_{2, (Q \setminus K) \cap M_i, g}^2 < \varepsilon^2; \quad (3.1)$$

- (iii) For each nonsingular terminal index $s \in S \cap M_i$, a single-valued lift on a collar of ∂T_s has boundary oscillation $A_i(T_s)$ satisfying

$$\left(\sum_{\substack{s \in S \cap M_i \\ s \neq p_i}} [A_i(T_s) - a_i(s)]_+^2 \right)^{1/2} < \varepsilon. \quad (3.2)$$

Here $[x]_+ = \max\{x, 0\}$. For a real coordinate the restriction $s \neq p_i$ is omitted.

- (iv) A periodic coordinate still adds 1 under its positive deck translation. Its differential has period 1 around the terminal disk indexed by its singular point, and period 0 around every other terminal disk in its mask. Here a positive circuit around a terminal disk is counterclockwise around that disk.

The core K_0 may contain any prescribed compact subsets of G . In particular the conformal structure on $\{|z| \geq R\}$ can always be retained. Terminal disks can be required to lie in arbitrarily small prescribed dust neighborhoods of their indices.

Proof. We first replace the coordinates outside a large core by shifted staircases. Compatibility will provide disks on whose collars at most one staircase varies. We can then fill those disks so that the extended differentials have rank at most one, and choose a metric from these differentials.

We regard du_i as zero wherever its mask is absent; every use of this convention is away from mask boundaries. Fix $\eta > 0$, to be chosen in terms of ε at the end.

Step 1: random staircases. Choose a smooth core K containing K_0 , all mask boundaries, and the exterior $\{|z| \geq R\}$ in its interior. Enlarge it until the original energies outside K are so small that

$$\sum_{i < j} \int_{G \setminus K} |du_i \wedge du_j| \leq \sum_{i < j} \|du_i\|_{2, G \setminus K} \|du_j\|_{2, G \setminus K} < \frac{\eta}{8}. \quad (3.3)$$

The mask intersections are implicit in these integrals. Take $\chi \in C^\infty(G, [0, 1])$ equal to 0 on a neighborhood of K and equal to 1 outside a larger compact set.

The staircases will vary only near a finite grid of levels modulo one. Let $h = 1/N$, where N is a positive integer. Independently for each i , choose θ_i uniformly in $[0, h)$. For $0 < \delta < h/4$ take a smooth nondecreasing staircase $P_{i, \delta}$ with

$$P_{i, \delta}(x + h) = P_{i, \delta}(x) + h, \quad |P_{i, \delta}(x) - x| \leq h,$$

whose derivative is supported within distance δ of $\theta_i + h\mathbb{Z}$. It is obtained by translating one fixed staircase with an h -periodic derivative of integral h over a period. Therefore

$$\mathbb{E} P'_{i,\delta}(x) = 1 \quad \text{for every } x. \quad (3.4)$$

Because $1 = Nh$, it also satisfies $P_{i,\delta}(x+1) = P_{i,\delta}(x) + 1$.

Use the cutoff to join each staircase to the original coordinate:

$$v_{i,\delta} = u_i + \chi(P_{i,\delta}(u_i) - u_i). \quad (3.5)$$

The formula is interpreted in local branches and respects integer translations, so it is consistent for periodic coordinates. Set

$$q_{i,\delta} = 1 - \chi + \chi P'_{i,\delta}(u_i), \quad r_{i,\delta} = P_{i,\delta}(u_i) - u_i.$$

Then $q_{i,\delta} \geq 0$, $|r_{i,\delta}| \leq h$, and

$$dv_{i,\delta} = q_{i,\delta} du_i + r_{i,\delta} d\chi.$$

Equation (3.4) and independence for distinct labels give $\mathbb{E} q_{i,\delta} = 1$ and $\mathbb{E}(q_{i,\delta} q_{j,\delta}) = 1$ for $i \neq j$. Consequently the wedge sum

$$W_\delta = \sum_{i < j} \int_{G \setminus K} |dv_{i,\delta} \wedge dv_{j,\delta}|$$

satisfies

$$\mathbb{E} W_\delta \leq \sum_{i < j} \int_{G \setminus K} |du_i \wedge du_j| + (m-1)h \sum_i \int_{\text{supp } d\chi} |d\chi| |du_i| da. \quad (3.6)$$

The last integrals are finite because $\text{supp } d\chi$ is compact in G . With K and χ fixed, choose h small enough that the right side is less than $\eta/4$, uniformly in δ .

Step 2: choosing the grids and terminal indices. For the random shifts above, put

$$F_i^{\text{grid}} = \bigcup_{k=0}^{N-1} H_i(\theta_i + kh).$$

Each grid is compact. In addition to F , retain as terminal indices all flagged points lying in the grids of two distinct labels:

$$S = F \cup \bigcup_{i < j} (\Sigma \cap F_i^{\text{grid}} \cap F_j^{\text{grid}}), \quad J = |S|.$$

At a nonsingular dust point, a band of length at most $a_i(s)$ meets the translated lattice with probability at most $\min(1, a_i(s)/h)$. The same bound holds at the singular point because there $a_i(s) = 1$. Thus, by independence,

$$h^2 \mathbb{E} J \leq h^2 |F| + \sum_{i < j} \sum_{s \in \Sigma} \min(h, a_i(s)) \min(h, a_j(s)) \longrightarrow 0 \quad (h \downarrow 0). \quad (3.7)$$

Each summand tends to zero and is bounded by $a_i(s)a_j(s)$, whose sum is finite by Cauchy–Schwarz. Dominated convergence proves the limit. The same estimate gives $J < \infty$ almost surely for each h .

Decrease h , if necessary, so that $h^2 \mathbb{E} J < \eta^2/4$ as well. Fix a deterministic sequence $\delta_k \downarrow 0$ with $\delta_k < h/4$. Fatou's Lemma and Equation (3.6) give

$$\mathbb{E}(\liminf_{k \rightarrow \infty} W_{\delta_k}) < \eta/4.$$

Markov's inequality shows that a set of positive probability satisfies both

$$\liminf_{k \rightarrow \infty} W_{\delta_k} < \eta, \quad h\sqrt{J} < \eta. \quad (3.8)$$

We also require compatibility (P) for every choice of one lattice level for each of any collection of distinct labels. Each such tuple has an absolutely continuous distribution in its level variables, and there are only finitely many choices. Hence this additional requirement has probability one. Fix grids satisfying it and Equation (3.8).

The grids and their finite set S are now fixed. Choose terminal disks about the points of S , with pairwise disjoint closures, in the region where $\chi = 1$. They contain no other point of S , lie in the prescribed dust neighborhoods, and avoid all mask boundaries. For every nonsingular coordinate present at s , its values in a sufficiently small neighborhood of s , in an appropriate local lift, lie in its band enlarged by h at each end. We therefore choose T_s so that

$$\text{osc}_{\partial T_s} u_i \leq a_i(s) + 2h.$$

The bound $|P_{i,\delta}(x) - x| \leq h$ gives, simultaneously for every δ ,

$$\text{osc}_{\partial T_s} v_{i,\delta} \leq a_i(s) + 4h. \quad (3.9)$$

No real oscillation bound is required at a singular index. The terminal disks are now fixed, while the staircase width can still decrease.

Step 3: filling disks whose boundaries meet at most one grid. We claim that the remaining dust has a finite cover by disks, with pairwise disjoint closures and disjoint from the terminal disks, such that the boundary of each new disk meets at most one F_i^{grid} . All these disks can be placed where $\chi = 1$, inside the prescribed dust neighborhoods, and away from mask boundaries.

First observe that if $s \in B \cap F_j^{\text{grid}}$ is not a terminal index, then s is a singleton component of

$$B \cup \bigcup_{i < j} F_i^{\text{grid}}. \quad (3.10)$$

If s is flagged, it does not belong to any lower grid, by the choice of S . If s is free, each coordinate defined at s has a unique trace. Thus, near s , only one level from each of the grids occurring at s can be present; the other levels are closed and miss s . If a nondegenerate component of the lower grid union contained s , intersect that component with a sufficiently small closed neighborhood of s . Boundary bumping makes the component of this intersection through s nondegenerate. It uses only the lower levels present at s , contradicting the chosen instance of compatibility (P). Lemma 2.2(i) shows that adjoining B cannot create another nondegenerate continuum through s . This proves the singleton assertion in both cases.

Process the labels in decreasing order. At the stage for j , the remaining dust is a compact relatively clopen subset of B , disjoint from the already chosen closed disks and from every higher grid. Use Lemma 2.2 to cover it by finitely many disjoint small preliminary disks whose closures avoid these obstacles. In each such disk V , the remaining points of $F_j^{\text{grid}} \cap B$ form a compact set. Each is a singleton component of the closed set in Equation (3.10); hence Lemma 2.2 encloses it in a smaller disk with closure in V and boundary avoiding that closed set.

Take finitely many such disks covering the compact target. To merge them while preserving boundary avoidance, perturb their finitely many smooth boundaries into general position within the open complement of the obstacle, take the outer contours of their union, and smooth the corners there. Fill bounded complementary regions and discard nested contours. Every resulting

disk stays in V , because the complement of V is connected and contains infinity. The resulting closures are pairwise disjoint, their interiors still cover the target, and their boundaries avoid all lower grids. They also avoid the higher grids, since their closures lie in V . Add these disks to the chosen collection and continue. After the last label, cover the dust still remaining by disks avoiding all the grids. This proves the claim. Call all disks chosen in this step *filling disks*.

Step 4: smooth fillings of rank at most one. The finite collection of terminal and filling disks covers B in its interiors. Its complement is therefore a compact subset of G . On each filling boundary, every coordinate except possibly one avoids its grid. Choose connected collars, compact in G , on which this avoidance still holds and $\chi = 1$. Every singular point lies in its assigned terminal disk, so the coordinates have single-valued lifts on each filling collar. When the collar avoids F_i^{grid} , a real lift has connected image in one interval complementary to the level lattice $\theta_i + h\mathbb{Z}$. Compactness of the collar then makes its staircase constant for all sufficiently small δ . Call these constant coordinates inactive on the collar. There are only finitely many collars, so one width bound works for all of them.

By Equation (3.8) we may choose one of the δ_k small enough for all these collar conditions and with $W_{\delta_k} < \eta$. Fix this width and suppress it from the notation. Retain v_i on the complement of the selected disks. Across each filling disk extend the inactive coordinates constantly. Extend the one possibly varying coordinate from an inside collar by a smooth cutoff, making it constant on a smaller interior disk. These extensions agree with the retained coordinates on full collars and are therefore smooth. Their differentials have rank at most one at every point of a filling disk, so

$$du_i^* \wedge du_j^* = 0 \quad \text{on every filling disk.} \quad (3.11)$$

Here u_i^* denotes the resulting coordinate on the completed bordered surface Q .

To check the periods, pull back by π the cyclic cover of the punctured mask. This cover is trivial on each filling disk. Perform the extension on one sheet and use its integer translates on the others. Equation (3.5) respects the same translations, so the completed coordinate has its original additive period. Under π , a counterclockwise circuit about T_{p_i} winds once about p_i , while a circuit about any other terminal disk in the mask has winding number zero. Thus the periods are exactly those asserted in the theorem.

Step 5: choosing a metric. The coordinates are now extended, with small wedge integrals outside K and zero wedge products in the filling disks. It remains to turn this control into small energy in one smooth metric. Let g_0 be the spherical metric and, outside K , form the smooth positive semidefinite tensor

$$A = \sum_{i: M_i \text{ present}} du_i^* \otimes du_i^*.$$

The sum at a point includes the masks containing that point. All mask boundaries lie inside K , so this tensor is smooth where it is used. The differentials are single-valued even for periodic coordinates. For $\tau > 0$ put $g_\tau = A + \tau g_0$. In a g_0 -orthonormal frame, let $\lambda_1, \lambda_2 \geq 0$ be the eigenvalues of A . The sum of the energy densities of the coordinates in g_τ , expressed relative to da_{g_0} , is

$$\text{tr}(g_\tau^{-1} A) \sqrt{\det g_\tau} = \frac{2\lambda_1\lambda_2 + \tau(\lambda_1 + \lambda_2)}{\sqrt{(\lambda_1 + \tau)(\lambda_2 + \tau)}} \longrightarrow 2\sqrt{\lambda_1\lambda_2}. \quad (3.12)$$

In particular the limit is zero when A has rank at most one. The Cauchy–Binet identity gives

$$\det A = \sum_{i < j} |du_i^* \wedge du_j^*|_{g_0}^2.$$

The transferred differentials are bounded on the finitely many smooth compact pieces of the completed surface. The densities in Equation (3.12) therefore have a common integrable bound for $0 < \tau \leq 1$: each term $\lambda_1 \sqrt{(\lambda_2 + \tau)/(\lambda_1 + \tau)}$ is at most $\sqrt{\lambda_1(\lambda_2 + 1)}$, and similarly for the other term. Dominated convergence is therefore applicable. By Equation (3.11) and the choice of the width,

$$\lim_{\tau \downarrow 0} \sum_i \|du_i^*\|_{2,(Q \setminus K) \cap M_i, g_\tau}^2 = 2 \int_{Q \setminus K} \sqrt{\det A} da_{g_0} \leq 2W_\delta < 2\eta. \quad (3.13)$$

Choose τ so that the total energy outside K in g_τ is less than 3η . With this regularization fixed, interpolate between g_0 and g_τ in a thin exterior collar of ∂K , keeping g_0 on a neighborhood of K . The interpolated metrics are uniformly positive definite and their energy densities are bounded on the collar. We may therefore choose its thickness so that the interpolation adds less than η to the total energy. Extend the resulting metric smoothly across each terminal cap, first across its collar and then by interpolation with g_0 inside the cap. This does not change the metric on Q .

The tail energy is now less than 4η . Moreover, Equation (3.9) gives for each i

$$\left(\sum_{s \neq p_i} [A_i(T_s) - a_i(s)]_+^2 \right)^{1/2} \leq 4h\sqrt{J} < 4\eta.$$

Taking $\eta < \min(\varepsilon^2/4, \varepsilon/4)$ establishes Equations (3.1) and (3.2). All alterations occurred outside a neighborhood of K , completing the proof. $\square$

We will also need energy control away from a prescribed part of the core. Exact agreement on K and the tail estimate give, for every measurable $E \subset K \cap M_i$,

$$\|du_i^*\|_{2,(Q \cap M_i) \setminus E, g}^2 \leq \|du_i\|_{2,(G \cap M_i) \setminus E}^2 + \varepsilon^2. \quad (3.14)$$

Both this estimate and the boundary oscillation estimates are unchanged by a conformal uniformization of Q° .

3.2 Circle-domain completions and compactness

We now apply finite uniformization to these completions and obtain a subsequential limit on the original domain. Keeping a fixed exterior region unchanged will give one normalization and one bound for all the finite maps.

Extend the metric of Theorem 3.1 across the terminal caps, as in its proof. Compact genus-zero uniformization identifies this metric sphere conformally with $\widehat{\mathbb{C}}$ [1, Theorem 10-3]. Classical finite circle-domain uniformization then maps Q° conformally onto a circle domain; only the finite-connectivity theorem is used here [12]; see also [3, Theorem 15.7.9]. Postcompose by a Möbius transformation sending the image of infinity to infinity, and then by an affine map, to obtain

$$f_Q(z) = z + O(1/z). \quad (3.15)$$

These postcompositions preserve roundness.

In the target coordinate w we use Euclidean length $|dw|$ and Euclidean area; on the source we continue to use physical spherical measures. Conformal invariance makes the Dirichlet energy norms agree.

Fix a countable compatible list and a countable set of dust points, called *marks*, containing all its singularities. Choose an enumeration of the underlying countable set of labels. The enumeration

is used only to choose increasing finite subsets; each subset is transferred in the order inherited from the original well-order. It need not be an initial segment of that well-order. This distinction allows a list of any countable ordinal type, including orders greater than ω . Apply Theorem 3.1 to these finite ordered sublists and increasing finite sets of marks. Prescribe cores containing an exhaustion of G , let the errors tend to zero, and place all terminal disks in members of dust covers whose mesh tends to zero. Denote the normalized map of the n th completion by f_n .

Each f_n is defined on its completed physical surface. The conformal structure there can differ from the original structure outside the retained core. For every fixed compact subset of G , the maps are eventually holomorphic and injective on a neighborhood of that subset in the original structure. Each fixed coordinate is eventually transferred, agrees with its original values on exhausting cores, and has errors tending to zero in Equations (3.1), (3.2), and (3.14).

Every fixed mark b eventually indexes a terminal disk shrinking to b in X . Write $D_{n,b}$ for its round complementary disk in the target. A finite completion may also have terminal indices outside the fixed countable set of marks; these additional indices may depend on n .

Lemma 3.2 (Normalized compactness). *There is a constant L , depending only on R , with the following properties. For every completion just described, all complementary disks of its normalized circle domain and all images of physical points in its domain with $|z| \leq 4R$ lie in $\{|w| < L\}$. One may take $L = 5R$. After passage to a subsequence, f_n converges locally uniformly on G to a normalized univalent map f . Simultaneously, for every fixed mark b , the disks $D_{n,b}$ converge in the Hausdorff metric to a closed round disk D_b , possibly a singleton.*

The same bound L holds for the bounded complementary components and the images of $G \cap \{|z| \leq 4R\}$ for every normalized univalent map on G .

Proof. The exterior $\{|z| \geq R\}$ retains its original conformal structure. Write

$$\frac{f_n(R\zeta)}{R} = \zeta + \sum_{j \geq 1} c_{n,j} \zeta^{-j}, \quad |\zeta| > 1.$$

The exterior univalent area theorem gives $\sum_{j \geq 1} j |c_{n,j}|^2 \leq 1$ [1, Theorem 5-1]. Hence, for $|\zeta| = r > 1$,

$$\left| \frac{f_n(R\zeta)}{R} - \zeta \right| \leq \left(\sum_{j \geq 1} \frac{r^{-2j}}{j} \right)^{1/2} = \sqrt{-\log(1 - r^{-2})}. \quad (3.16)$$

For $r \geq 2$ the right side is at most $\sqrt{\log(4/3)} < 1$.

The image of $\{|z| > 2R\} \cup \{\infty\}$ is the unbounded side of the Jordan curve $f_n(\{|z| = 2R\})$. Injectivity therefore puts all other image points and all complementary disks on its bounded side, which lies in $\{|w| < 3R\}$ by Equation (3.16). For $2R \leq |z| \leq 4R$ that equation gives $|f_n(z)| < 5R$. This proves the uniform bound. The same argument applies without change to any normalized univalent map on G .

On every relatively compact connected subdomain of $G \setminus \{\infty\}$, the maps are eventually holomorphic, injective, and uniformly bounded. Diagonal normal-family compactness therefore gives a locally uniform holomorphic limit. The exterior estimate preserves its Laurent normalization, so the limit extends across infinity as $f(z) = z + O(1/z)$ and is nonconstant. Hurwitz's theorem for limits of injective holomorphic maps, applied on each of these subdomains, makes the limit injective on all of G .

The centers and radii of the marked disks lie in a fixed compact parameter set. A further diagonal subsequence makes them converge for all countably many marks. Convergence of centers and nonnegative radii is precisely Hausdorff convergence of the disks, with radius zero giving a singleton. $\square$

3.3 Identification and trapping of the ends

We have obtained a univalent limit and round disks at the marked ends. To compare those disks with the full complement of the image, we identify each complementary component by nested source collars.

For a normalized univalent map f on G and a smooth Jordan curve $C \subset G$ enclosing a dust disk, write $\mathcal{Q}_f(C)$ for the filled closed bounded Jordan disk with boundary $f(C)$.

Lemma 3.3 (End components). *Let f be a normalized univalent map on G . For $b \in B$ set*

$$K_b(f) = \bigcap_{j \geq 1} \mathcal{Q}_f(C_j(b)). \quad (3.17)$$

These sets are exactly the complementary components of $f(G)$, with one component for each $b \in B$. They are nonempty continua, and

$$K_b(f) \subset \text{int } \mathcal{Q}_f(C_j(b)) \quad \text{for every } j.$$

For a sequence and marked disk limits as in Lemma 3.2,

$$D_b \subset K_b(f). \quad (3.18)$$

In a single finite completion, each terminal boundary corresponds to exactly one complementary round disk. Points of the circle domain approaching that disk have preimages approaching the corresponding terminal boundary as a set. Thus the cluster oscillation of a transferred coordinate at a nonsingular disk is at most its boundary oscillation $A_i(T_s)$, in a single-valued lift on a collar.

Proof. We first identify the components for an arbitrary normalized univalent f , then treat terminal boundaries in a finite completion and pass their marked disks to the limit.

Let K_j be a core from the fixed exhaustion. It is the closed exterior of finitely many disjoint dust disks and contains infinity in its interior. Its image under f is a compact embedded bordered surface whose boundary curves are the images of the source boundary curves. Orientation fixes the image side of each boundary in a collar. The connected interior of the core, which contains infinity, must remain on that exterior side. Jordan separation therefore identifies $f(K_j)$ with the closed exterior of the corresponding pairwise disjoint filled bounded Jordan disks.

Under refinement of the cores these image disks are nested in exactly the same way as their source dust disks. Strict nesting in the source gives

$$\mathcal{Q}_f(C_{j+1}(b)) \subset \text{int } \mathcal{Q}_f(C_j(b)).$$

Their intersection is a nonempty compact continuum, strictly inside each earlier image disk. It misses $f(G)$, because every point of G lies in the interior of some exhaustion core, while the intersection lies outside all image cores.

Conversely, a point outside $f(G)$ lies inside exactly one image dust disk at every exhaustion level. Refinement makes the corresponding source disks nested. Their mesh tends to zero, so they determine a unique point $b \in B$, and the target point belongs to $K_b(f)$. A connected subset of the complement cannot meet two disjoint image disks at the same level. Thus the component containing a point of $K_b(f)$ lies in every disk in Equation (3.17). The intersection is itself connected, so it is the whole component. Distinct dust points are separated at some exhaustion level. This proves the bijection.

For a finite completed surface, use cores bounded by successively smaller collars of its terminal boundaries. The same argument gives one end per terminal boundary; its nested image disks intersect in the corresponding round complementary disk. Fix such a disk D and a source collar curve C . The disk D is compactly contained in the bounded side of the image of C . Hence target points whose distance from D tends to zero eventually lie on that side, and their preimages lie between C and the terminal boundary. Taking arbitrarily small collars shows that the preimages approach this boundary as a set.

The transferred coordinate is smooth up to the terminal boundary, so its cluster values lie in its boundary range. At a nonsingular terminal index its period is zero, and a single-valued lift can be used on the whole collar. This gives the cluster oscillation bound without a boundary extension of the uniformizing map.

Finally fix a mark b and an exhaustion curve $C_j(b)$. For all sufficiently large n this curve is retained on the core and the terminal disk indexed by b lies on its bounded source side. Jordan separation in that finite completed surface gives

$$D_{n,b} \subset \text{int } \mathcal{Q}_{f_n}(C_j(b)).$$

Uniform convergence on the curve and stability of winding number put every Hausdorff limit point of these disks in $\mathcal{Q}_f(C_j(b))$. Intersecting over j proves Equation (3.18). For strict trapping inside the image of a fixed collar, first use $C_{j+1}(b)$ and then its compact containment in the bounded side of $f(C_j(b))$. $\square$

The later selections will enlarge the coordinate list, flags, and marks. Every countable compatible enlargement still admits the construction above. Each fixed coordinate and each fixed mark are eventually included in its diagonal transfers, so all their estimates and the trapping conclusions persist.

4 Tagged paths and compatible barriers

Fix a countable ordered list of coordinates satisfying (P), with flag set Σ . Throughout this section the list is fixed. We seek a bounded smooth real coordinate that vanishes on a compact testing set in G and equals one on a collar about a chosen dust point. The coordinate must remain compatible with the prior list. A lower bound on path costs will supply the separation, while the norms of the density and boundary tolls will control the barrier's size.

4.1 Endpoint data and barriers

The endpoint data specify the two sets to be separated. Gates optionally restrict where paths cross the prescribed collars; the controlled energy then omits a permitted penalty region. In the ungated case this region is empty.

Definition 4.1 (Endpoint data and a barrier). An endpoint problem consists of a point $b \in B$, a nested sequence of prescribed dust disks $U_j = U_j(b)$ with boundaries C_j , and a compact anchor $E \subset G$ with nonempty interior, disjoint from all $\overline{U_j}$. A compact set $E_* \subset \text{int } E$ is also specified. The collar sequence may start at any depth in the fixed exhaustion.

There may additionally be *gates*: a fixed normalized univalent map f_0 on G , a bounded open Euclidean ball W_2 , and a closed ball $W_1 \subset W_2$. Every hit of every C_j must then lie in $f_0^{-1}(\overline{W_2})$. In this case a *penalty region* is any relatively compact open $O \subset G$ such that

$$\overline{O} \cap E = \emptyset, \quad f_0(\overline{O}) \cap W_1 = \emptyset.$$

In the absence of gates put $O = \emptyset$.

A barrier is a bounded smooth real coordinate u on G , with finite total energy and budget $a_u \in \ell^2(B)$, such that $u = 0$ on E_* , $u = 1$ on some C_l , and appending u preserves (P) after a countable enlargement of the flag set. Its controlled size is

$$\|du\|_{2,G \setminus O} + \|a_u\|_{\ell^2(B)}.$$

Energy on O is not included in this size, but must still be finite.

4.2 Path length, tags, and residual contacts

Every path has parameter interval $[0, 1]$ and is continuous as a map $\gamma : [0, 1] \rightarrow X$. On the open set $\gamma^{-1}(G)$ it is a path in the physical sphere. Its *domain-part length* is the sum of its physical spherical lengths on the components of this open set, where the length on an open component is obtained by exhaustion by compact subintervals. Only paths with finite domain-part length will be used. Their arclength occupation measure on G is denoted by ν_γ ; it counts length with parameter multiplicity. When an integral is written over parameter times, ds means the path's domain-part arclength measure on those times, not Lebesgue measure dt ; its pushforward to G is ν_γ .

We call a pair (g, m) a *metric* when $g \in L^2(G, da)$ is nonnegative Borel, possibly with infinite values on a null set, and $m \in \ell^2(B)$ is nonnegative. The use of a domain density together with square-summable boundary weights is in the tradition of Schramm's transboundary extremal length [19]. Here a boundary toll is charged at every visit time. This convention will be used in the concatenation and intersection arguments. The true-path cost is

$$\mathcal{L}_{g,m}(\gamma) = \int_G g d\nu_\gamma + \sum_{r \in \gamma^{-1}(B)} m(\gamma(r)). \quad (4.1)$$

The second term counts *times*, not merely distinct image points. As elsewhere, its value is the supremum of finite subsums. Thus a positive toll at a point visited infinitely many times gives infinite cost. We write $\mathcal{L}_{g,m}^-$ when the time 1 is omitted from the toll sum, and $\mathcal{L}_{g,m}^\circ$ when both times 0 and 1 are omitted. These omissions do not affect the arclength term.

Arclength on the open domain times is atomless. Its occupation dominates $\mathcal{H}_{\text{sph}}^1$ on the image in G . If γ is injective and its image in G has finite physical $\mathcal{H}^1$ measure, then its domain-part length is finite and its occupation is precisely that image measure. To check these statements it suffices to exhaust the open domain times by compact intervals. On each such interval one has the ordinary facts for continuous rectifiable curves. For an injective curve the partition sums are bounded by its image length, because the intervening subarcs have disjoint interiors. In particular, all uses of curve coarea below concern ordinary rectifiable portions in the physical domain, irrespective of the behavior at dust times.

We next record portions of a path that travel inside finite unions of prior level sets. A tag will specify such a union and the parameter interval on which the path follows it. The dust contacts outside these intervals will be subject to separate restrictions.

A *type* is a nonempty finite list $\iota = (i_1, \dots, i_d)$ of prior labels, with repetitions allowed. Write

$$H_\iota(\mathbf{t}) = \bigcup_{j=1}^d H_{i_j}(t_j), \quad \mathbf{t} = (t_1, \dots, t_d) \in \mathbb{T}^d.$$

There are countably many types. For each type fix a Haar-null Borel set $N_t \subset \mathbb{T}^d$ of excluded tuples; it may be empty. A path has countably many optional *tag slots*. An active slot specifies a type, a tuple outside its excluded set, and an interval (a, b) with $0 \leq a < b \leq 1$, such that γ is nonconstant on (a, b) and its image there is contained in $H_t(\mathbf{t})$. The entire interval $(0, 1)$ is permitted, and different tags may overlap. For a tagged path $\Gamma = (\gamma, \theta)$, define the compact set of residual dust times by

$$Z_\Gamma = \gamma^{-1}(B) \setminus \bigcup_{\text{active slots}} (a, b).$$

The additional data defining a path system are as follows:

- a finite atomless Borel measure λ on B , possibly zero;
- a countable collection $\mathcal{Q}$ of earlier probability laws on paths, for each of which

$$m_\xi(s) = \mathbb{P}_\xi\{s \in \gamma([0, 1])\}, \quad s \in B, \quad \text{satisfies} \quad m_\xi \in \ell^2(B);$$

- increasing finite sets $P_{\xi,n}$ exhausting $\{m_\xi > 0\}$, and increasing finite sets $P_{\Sigma,n}$ exhausting Σ .

Finite exhaustions may stabilize. Laws carrying their own tags are evaluated on their path marginals in the displayed definition.

The residual image must have zero λ measure. Its incidences with earlier path laws and prior levels will be bounded by the corresponding point weights, summed over residual times, on each rational test interval and after each prescribed finite deletion.

Definition 4.2 (An allowed tagged path). A finite-domain-length tagged path is *allowed* if all of its tags obey their tuple exclusions, if

$$\lambda(\gamma(Z_\Gamma)) = 0, \tag{4.2}$$

and if the following inequalities hold for every closed interval $I \subset [0, 1]$ with rational endpoints, including $I = [0, 1]$:

$$\begin{aligned} \mathbb{P}_\xi\{\text{the sampled path visits } \gamma(Z_\Gamma \cap I) \setminus P_{\xi,n}\} \\ \leq \sum_{\substack{r \in Z_\Gamma \cap I \\ \gamma(r) \notin P_{\xi,n}}} m_\xi(\gamma(r)) \quad (\xi \in \mathcal{Q}, n \in \mathbb{N}), \end{aligned} \tag{4.3}$$

$$\begin{aligned} |\{v \in \mathbb{T} : H_i(v) \cap (\gamma(Z_\Gamma \cap I) \setminus P_{\Sigma,n}) \neq \emptyset\}| \\ \leq \sum_{\substack{r \in Z_\Gamma \cap I \\ \gamma(r) \notin P_{\Sigma,n}}} a_i(\gamma(r)) \quad (i \text{ a prior label}, n \in \mathbb{N}). \end{aligned} \tag{4.4}$$

An untagged path is called allowed if it admits such tags. Unless otherwise indicated, path spaces and probability laws in the arguments below retain the tags as part of their codes.

For a singleton image $\{s\}$ the left sides of Equations (4.3) and (4.4) are bounded by $m_\xi(s)$ and $a_i(s)$, respectively. Countable subadditivity therefore shows that all the constraints hold whenever the residual image is countable. In particular, ordinary finite-length paths in G are allowed. A simple arc of finite domain-part length in an entire permitted level union is also allowed: give it the one tag $(0, 1)$, leaving at most its two endpoints in the residual image.

4.3 The compatible barrier theorem

With the endpoint data and allowed path class specified, the construction has the following quantitative form.

Theorem 4.3 (Compatible barrier lemma). *Use any fixed path system of Definition 4.2 and any endpoint data of Definition 4.1. Suppose $g \in L^2(G)$ is nonnegative Borel and bounded below by a positive constant, and $m \in \ell^2(B)$ is nonnegative. Assume that every allowed path from E to b which first reaches b at time 1 and obeys the gates, if present, satisfies*

$$\mathcal{L}_{g,m}^-(\gamma) \geq 1.$$

Then for every $\varepsilon > 0$ there is a barrier u , and an allowed penalty region O when gates are present, such that

$$\|du\|_{2,G \setminus O} + \|a_u\|_{\ell^2(B)} \leq C_{\text{bar}}(\|g\|_2 + \|m\|_{\ell^2(B)}) + \varepsilon, \quad C_{\text{bar}} = 8. \quad (4.5)$$

One may take $0 \leq u \leq 1$, $a_u = 4m$, and enlarge the flags only by $\{m > 0\}$. The numerical constant is independent of the prior list, earlier laws, atomless measure, null tuple exclusions, endpoint data, and positive lower bound on g .

4.4 Cutting, joining, and coding paths

We prove the path properties needed for the theorem before constructing the coordinate. Cutting and joining must preserve the residual-contact bounds; Borel coding will ensure measurability of the cost infimum.

Lemma 4.4 (Restrictions and countable concatenations). *Allowed paths are stable under reversal, affine reparametrization, and restriction to a closed parameter subinterval. More generally, suppose a continuous path is made from countably many restrictions of allowed paths, possibly reversed, on disjoint open parameter intervals. Assume that the complement of those intervals contains only countably many times and that the resulting path has finite domain-part length. Then the path can be tagged so as to be allowed.*

Proof. For each of the two incidence inequalities, its left side is a countably subadditive set functional of the residual image, continuous under increasing unions. Its value on $\{s\}$ is bounded by the corresponding weight at s . Consequently the inequalities for rational closed intervals extend first to arbitrary open time intervals by an increasing exhaustion. They then extend to arbitrary closed intervals by treating the two endpoints separately.

Restrict every old tag to the interior of the retained piece. Keep the restriction if the path is nonconstant there, and discard it otherwise. The residual image newly created by the discarded tags is countable: each discarded restriction has one constant image point, and there are only countably many tags. This proves the restriction claim, because countable image additions satisfy the incidence bounds by singleton estimates and have zero λ measure. Reversal and affine reparametrization are treated by pulling back the time intervals.

For the countable concatenation, keep these restricted tags in the piece interiors. The residual times fall into three groups: retained old residual times, times exposed by discarded constant tags, and junction or limit times. Fix a test interval I and a finite deletion set P . For each open piece interval J , let R_J be the pullback of the old residual times in $J \cap \text{int } I$. These times are still residual, because every new tag is a restricted old tag. Set

$$A_J = (Z_{\text{new}} \cap J \cap \text{int } I) \setminus R_J.$$

The image of A_J is countable: every such time belongs to a discarded constant-tag restriction. The remaining residual times T consist of junction or limit times, together with at most the two endpoints of I , and hence are countable. The sets R_J , A_J , and T are pairwise disjoint as sets of times. The old bounds on open intervals apply to each R_J . If $A = T \cup \bigcup_J A_J$, its countable image satisfies, for either incidence functional $\mathcal{C}$ and its corresponding weight w ,

$$\mathcal{C}(\gamma(A) \setminus P) \leq \sum_{s \in \gamma(A) \setminus P} w(s) \leq \sum_{\substack{r \in A \\ \gamma(r) \notin P}} w(\gamma(r)).$$

Apply the union bound on image sets and sum the old bounds with this last one. All charged times are disjoint, so their sum is bounded by the required right side for the new path, even when their images overlap. The same decomposition proves the atomless condition. Every retained tag has its old tuple and hence still avoids the prescribed null set.

This proof explains the countability assumption on the extra times: countably many open piece intervals alone could leave a Cantor set of uncontrolled times. In the collar concatenations below the intervals accumulate only at explicitly specified dust endpoints, so the stated assumption holds. $\square$

To optimize over paths and place probability laws on their codes, we also need the path constraints and costs to be Borel.

Lemma 4.5 (Borel coding of paths and costs). *For fixed path-system data, the allowed tagged codes form a standard Borel space. On this space the domain-part length and the arclength occupation measure are Borel. For every fixed nonnegative Borel $g \in L^2(G)$ and fixed nonnegative $m \in \ell^2(B)$, all three costs $\mathcal{L}_{g,m}$, $\mathcal{L}_{g,m}^-$, and $\mathcal{L}_{g,m}^\circ$ are Borel, with values in $[0, \infty]$. Endpoint restrictions, first arrival at a prescribed dust point at time 1, and hit or avoidance conditions for closed or relatively open subsets of prescribed compact collar curves are Borel.*

Proof. Tags and residual times. Start with $C([0, 1], X)$ and the countable product of the slot spaces. A slot is either inactive, or belongs to a countable disjoint union of spaces consisting of a type, a level tuple, and endpoints $0 \leq a < b \leq 1$. These are standard Borel spaces. Give the inactive and active alternatives separate components. Image containment for a tag can be tested at rational times in (a, b) , using the closed level graphs. Nonconstancy can be tested by two such rational times with different images. Both requirements, as well as the tuple exclusion, are Borel. In this ambient coding the residual-time graph is closed relative to the code space times $[0, 1]$, and in particular has compact time sections.

Incidence and visit counts. For Equation (4.3), introduce the sampled path η and two time variables r, u . The incidence in question is equivalent to

$$r \in Z_\Gamma \cap I, \quad \gamma(r) = \eta(u), \quad \gamma(r) \notin P,$$

for some $(r, u) \in [0, 1]^2$. Here P is finite. The last condition is a countable union of closed conditions $\text{dist}_X(\gamma(r), P) \geq 1/n$; for $P = \emptyset$ no condition is needed. Thus the witness sections are sigma-compact, and the Arsenin–Kunugui projection theorem for sigma-compact sections applies [10, Theorem 18.18]. For Equation (4.4), use instead the closed incidence relation $\gamma(r) \in H_i(v)$. The same argument applies. Integration against the fixed law ξ or Haar measure shows that the two left sides are Borel functions of the tagged code. The test $s \in \gamma(Z_\Gamma)$ is likewise Borel in (Γ, s) ; integration against λ gives the Borel atomless condition.

For a fixed point s , at least k visits in a specified closed residual time set can be tested with k time variables. Distinctness of those times is the union, over n , of the closed conditions

$|r_j - r_l| \geq 1/n$ for $j \neq l$. Projection again gives a Borel visit count $N_s \in \{0, 1, \dots, \infty\}$. Since every counting ℓ^2 weight has countable positive support,

$$\sum_{r \in \gamma^{-1}(B)} m(\gamma(r)) = \sum_{s: m(s) > 0} m(s) N_s(\gamma)$$

is Borel. The same argument treats the right sides of the two constraints. Omitting one or both endpoint times is handled by exhausting the resulting open or half-open parameter intervals.

Physical arclength. For a rational closed interval J , the condition $\gamma(J) \subset G$ is open in path space. On this set the physical restriction depends continuously on the path, locally uniformly in a compact subset of G . Its length is the supremum of the partition sums, and rational partitions suffice by continuity. Hence the length is Borel. On finite-length restrictions, the integral of a bounded nonnegative continuous function f against arclength is the limit of sums

$$\sum_j f(\gamma(t_j)) \text{length}_{\text{sph}}(\gamma|_{[t_j, t_{j+1}]}) ,$$

over increasingly fine equal subdivisions. Uniform continuity of $f \circ \gamma$ and finiteness of total length justify this limit. The full domain-part integral is the supremum of sums over finite disjoint collections of rational closed intervals contained in the open domain times. Such collections exhaust all those times, and arclength has no atoms. This proves the Borel length assertion. A countable determining family of continuous functions on G also shows that $\gamma \mapsto \nu_\gamma$ is Borel into the space of finite Borel measures on G . Consequently $\int g d\nu_\gamma$ is Borel for every fixed nonnegative Borel g , by the usual monotone-class argument, including extended values.

The finite-length condition and all the constraints are therefore Borel, proving the assertion about the tagged space. Endpoint conditions are immediate. Avoidance of a fixed point on $[0, 1)$ can be tested on the compact intervals $[0, 1 - 1/n]$. Hit conditions for closed subsets of a compact collar have compact time sections; relatively open collar subsets are sigma-compact and give the same Borel conclusion. These observations prove the final assertion. $\square$

Remark 4.6. The preceding lemma concerns tagged codes. The set of untagged paths that *admit* tags is an analytic projection and need not be Borel. Likewise, cost measurability is asserted for each fixed metric (g, m) ; no Borel parametrization of all counting $\ell^2(B)$ weights is being asserted. All probability measures on analytic sets may be completed, as in Section 2.

4.5 Collar concatenation

The following lemma converts bounded-cost paths approaching dust endpoints into paths that reach them. It is the compactness substitute used twice in the barrier construction: assembling actual path pieces lets us retain the path restrictions through the stability lemma.

Lemma 4.7 (Collar concatenation). *Fix a path system and a metric (g, m) with $g \geq c_0 > 0$ on G . Let allowed paths γ_n , whose endpoints are in G , have costs converging to a finite number A .*

- (i) *If their initial and final endpoints converge to distinct points $p, q \in B$, then for every $\eta > 0$ there is an allowed path γ from p to q satisfying*

$$\mathcal{L}_{g,m}^\circ(\gamma) \leq A + \eta.$$

- (ii) *If their initial endpoints belong to a set $F \subset G$ and their final endpoints converge to $p \in B \setminus \overline{F}$, where closure is in X , there is an allowed path from a point of F to p with $\mathcal{L}_{g,m}^- \leq A + \eta$.*

The paths produced consist of countably many restricted original pieces and joining arcs on smooth collar curves in G . Every dust point avoided by the original pieces remains avoided away from the newly adjoined endpoint times. Any prescribed hit restrictions are also preserved when the joining collars are disjoint from their hit curves.

Proof. We take limits of collar hit positions and fragment costs, then join actual fragments with summable extra cost. Shrinking disks will give continuity at the new endpoint times.

Choice of finite-cost collars. About p and q choose strictly nested dust disks V_j and W_j with closures shrinking to their respective points and with disjoint outer closures. Their boundary curves $A_j = \partial V_j$ and $B_j = \partial W_j$ can be chosen to satisfy

$$\int_{A_j} g \, ds < \infty, \quad \int_{B_j} g \, ds < \infty.$$

Indeed each smooth boundary has a compact smooth annular collar in G , and Fubini in its smooth parallel coordinates gives finite integrals for almost every parallel curve. Arbitrarily small perturbations preserve all the required nesting and separation. For part (ii), choose only the collars about p , with the outer closed disk missing $\bar{F}$. When joining collars disjoint from additional hit curves are available, choose them. For the nested gate curves used below, the intervening annuli provide these choices.

Cuts and limiting budgets. Consider part (i). After passage to a subsequence the two endpoints of γ_n lie in V_n and W_n . For each fixed $j \leq n$, let $\alpha_{n,j}$ be its first hit of A_j and let $\beta_{n,j}$ be its last hit of B_j . The cuts are ordered as

$$\alpha_{n,n} \leq \cdots \leq \alpha_{n,1} < \beta_{n,1} \leq \cdots \leq \beta_{n,n}.$$

The strict middle inequality follows from disjointness of the outer closed disks. The initial fragment between $\alpha_{n,j+1}$ and $\alpha_{n,j}$ lies in $\bar{V}_j$, and the final fragment between $\beta_{n,j}$ and $\beta_{n,j+1}$ lies in $\bar{W}_j$. Retain also the middle fragment between $\alpha_{n,1}$ and $\beta_{n,1}$.

Every cut point belongs to G . Thus these fragments have disjoint interior time sets, no dust toll is duplicated at a cut, and arclength adds without endpoint atoms. A diagonal subsequence makes all the hit positions and all the fragment costs converge. Denote the limiting middle cost by c_* and the two families of limiting fragment costs by c_j^- and c_j^+ . Every finite partial sum is bounded by A , so

$$c_* + \sum_{j \geq 1} (c_j^- + c_j^+) \leq A. \quad (4.6)$$

Joining actual fragments. On each fixed collar, two points sufficiently close to the same limiting hit position can be joined by a collar arc whose g integral is arbitrarily small. This is absolute continuity of the integral of g along that finite-integral smooth circle. Assign summable positive error allowances, with total less than η , both to fragment costs and to all joining arcs. At each collar choose a neighborhood of its limiting hit position small enough for the assigned joining allowance. Select the middle fragment and every initial and final fragment from sufficiently far along its convergent subsequence so that its cost is at most its limit plus its allowance, and both endpoints are in the chosen collar neighborhoods. Adjacent selected fragments can then be joined with the assigned small costs.

Parametrize the resulting initial pieces on successive intervals accumulating at 0, the middle portion on a compact interior interval, and the final pieces on successive intervals accumulating at 1. The initial tails lie in shrinking $\bar{V}_j$ and the final tails in shrinking $\bar{W}_j$. Adding endpoint values p, q therefore gives a continuous path in X . Equation (4.6) and the assigned errors bound its cost

with the two endpoint tolls omitted by $A + \eta$. Since $g \geq c_0$, the sum of all domain-part lengths is finite. There are only countably many additional junction times. Lemma 4.4 makes the new path allowed.

For part (ii), use last hits of the collars about p . Keep one actual prefix, beginning at its original point of F , followed by the successive last-hit fragments. The same diagonal argument, budget inequality, and small collar arcs construct the required one-ended path. Finally, the construction uses only restrictions of original paths and arcs on the selected collars. This proves the asserted preservation of avoidance and hit restrictions. $\square$

4.6 Removing a newly introduced level of small jumps

Jumps inside small dust disks will control the eventual barrier's oscillation near the dust. We must show that adding these jumps preserves a positive cost for travel from an anchor to a collar.

A *jump level* consists of a finite pairwise disjoint cover of B by dust disks whose closures miss the anchor E and a specified collar C . Each disk has a positive toll, independent of the two endpoints of a jump. A jump joins any two domain points inside that disk. A finite chain alternates allowed true paths with jumps; all true-path endpoints and all jump endpoints belong to G . Its cost is the sum of true costs and jump tolls. A constant true path in G is allowed, so consecutive jumps cause no difficulty.

Lemma 4.8 (Removal of one shrinking jump level). *Fix a metric (g, m) with $g \geq c_0 > 0$, a compact anchor $E \subset G$, and a compact collar curve $C \subset G$. Suppose that, with finitely many fixed jump levels, every chain from E to C has cost at least $\beta > 0$. Fix $\rho, \tau > 0$ and $0 < \Delta < \beta$. For all sufficiently small meshes, adding any new jump level whose disk V has toll*

$$c(V) = \max \left\{ \rho, \sup_{\substack{s \in B \\ \text{dist}_X(s, V) < \tau}} m(s) \right\} \quad (4.7)$$

preserves the lower bound $\beta - \Delta$ for chains from E to C . Here each new disk meets B and its closure misses E and C .

Proof. In the contradiction argument, shrinking new jumps collapse to dust junctions. Collar concatenation replaces the adjoining path pieces, and the new-jump tolls pay for the junction visits.

Fixing the jump pattern and its limiting costs. Suppose instead that covers of mesh tending to zero admit chains of cost less than $\beta - \Delta$. All existing jump tolls and ρ are bounded below by a common positive number. The number of jumps in these chains is therefore uniformly bounded. After combining successive true paths and passing to a subsequence, fix the number of jumps, the pattern of old and new jumps, and the disk label at every old jump. The two endpoints of each new jump converge, after a further subsequence, to the same point $p_i \in B$: its disk meets B and has diameter tending to zero. All true-piece costs and all new-jump costs may also be assumed convergent. Since τ is fixed, the limiting cost of the new jump at p_i is at least $m(p_i)$.

Replacing the blocks between old jumps. Break each chain into blocks separated by old jumps. The anchors of a block are E , C , or the open disk of the adjacent old jump. We will replace the block by an allowed true path between its two anchors, with cost at most the sum of its limiting costs plus arbitrarily small error. Different blocks need not choose the same domain point inside an old jump disk: that old jump can still join their chosen endpoints at its fixed toll.

Suppose first that a block has new-jump limits $p_1, \dots, p_d$. Collapse each run of equal consecutive limits to one junction and discard the intervening true pieces. This only decreases the available budget; one of the jump budgets in the run still pays at least the required single toll. Between successive distinct junctions apply Lemma 4.7(i) to the corresponding true pieces. Their endpoints lie in G and converge to those junctions. This produces an allowed path between them at its limiting true-piece cost plus an arbitrary small error, with the two junction tolls temporarily omitted.

We must also attach the replacement to its two outer anchors, with endpoints in the domain. At an outer anchor A , if its adjacent junction p is outside $\bar{A}$, use the one-ended version of Lemma 4.7, reversing paths if necessary. If $p \in \bar{A}$, the anchor cannot be E or C , since these are compact subsets of G . It is therefore an old jump disk. Its boundary lies in G , so $p \in B \cap \bar{A}$ actually belongs to its interior. In this case discard the outer true piece and truncate the next constructed path at a domain point sufficiently close to p to lie in A . Such points exist arbitrarily close to p in the collar construction. More generally, a nonconstant arc cannot have an initial interval contained in the totally disconnected set B . The truncation is allowed by Lemma 4.4 and cannot increase the cost.

If a single remaining junction lies in both anchors, use instead a common domain point of their open intersection and the constant path. Such a point exists because G is dense in X . If precisely one anchor contains the sole remaining junction, construct the one-ended path from the other anchor to that junction and truncate its end on a sufficiently deep domain collar inside the first anchor. The truncation adds no dust toll, and reversal gives the required order. If the block has no new junctions, simply retain an actual true path with cost sufficiently close to its limit.

Accounting for the dust junctions and reconnecting the blocks. Concatenate the retained constructed paths at each remaining dust junction using a single shared time, without adding a pause there. Exactly one previously omitted toll is then incurred at that time. It is covered by the limiting new-jump cost, which is at least $m(p_i)$. Any additional internal visits to that point were already included in the true-piece costs. Infinite collar tails add only their specified endpoint times, and the construction uses countably many pieces in all. Its domain-part length is finite because $g \geq c_0$. Thus Lemma 4.4 supplies the required allowed true path replacing the block.

There are finitely many blocks. Choose their errors to have total less than $\Delta/2$ and reconnect them with the unchanged old jumps. The resulting old-level chain has cost at most $\beta - \Delta + \Delta/2 < \beta$, a contradiction. $\square$

4.7 Proof of the compatible barrier theorem

Proof of Theorem 4.3. Steps 1 and 2 reduce the endpoint hypothesis to a positive cost for reaching a fixed collar, with the relevant finitely many gates absorbed into a density on a penalty region. Steps 3–5 construct a separating function from chain costs, bound its topological bands, and smooth it with controlled energy while retaining those band bounds. Step 6 proves compatibility with the prior coordinates by applying the path-cost estimate to tagged arcs in their level unions.

Step 1: a finite collar already has positive cost. We claim that for some k , every allowed path from E to C_k obeying the gates through index k has full cost at least $1/2$. Otherwise choose violations with indices $k_n \rightarrow \infty$ and cut each at its first hit of C_{k_n} . The resulting paths do not hit b or any deeper prescribed collar. They consequently obey all the gates, including those with index greater than k_n , since those have no hits. Their terminal points converge to b . After taking a subsequence their costs converge to a number at most $1/2$.

Choose strictly intermediate smooth collars between the prescribed ones, with finite g integrals.

They avoid every prescribed hit curve. Apply the one-ended construction in Lemma 4.7 with error less than $1/4$. It gives an allowed path from E to b with final toll omitted and cost less than 1. The retained original pieces avoid b , the joining arcs lie in G , and all gates are preserved. Thus its first visit to b is at the final endpoint, contrary to hypothesis.

Step 2: replace finitely many gates by a penalty density. In the ungated case set $O = \emptyset$ and $g_1 = g$. With gates, form the compact set

$$F = \bigcup_{j=1}^k (C_j \setminus f_0^{-1}(W_2)) \subset G.$$

It is disjoint from E , and its image misses W_1 . Compact separation gives a relatively compact open set $O \subset G$ containing F whose closure misses E and whose image under f_0 misses W_1 . Choose a bounded nonnegative density h vanishing outside O such that every path from E to F has h -length at least 1. For example, if $F \neq \emptyset$, put

$$\delta = \text{dist}_{\text{sph}}(F, \widehat{\mathbb{C}} \setminus O) > 0, \quad h = \delta^{-1} \mathbf{1}_O.$$

The last passage from outside O to F is entirely in G and has physical length at least δ . If $F = \emptyset$, take $h = 0$ and $O = \emptyset$. Set $g_1 = g + h$. This density is in $L^2(G)$, remains bounded below by a positive constant, and agrees with g off O .

An allowed path from E to C_k which fails a gate through index k reaches F and has g_1 -cost at least 1. A path which obeys those gates has the $1/2$ lower bound from Step 1. Cutting first at C_k if necessary shows that every allowed path from E to C_k , without any gate restriction, now has full (g_1, m) -cost at least $1/2$.

Step 3: chain distance with progressively smaller jumps. Taking an infimum of path costs is the classical modulus-to-capacity construction; see [8, Proposition 2.17]. Here shrinking jumps will control the bands at every dust point. Choose positive losses Δ_j with $\sum_j \Delta_j < 1/8$. Starting with the true paths alone, add jump levels one at a time. At level j take $\rho_j, \tau_j \downarrow 0$ and a dust-disk cover of mesh tending to zero, with closures missing E and C_k . Lemma 2.2 permits these covers, and Lemma 4.8 permits choosing each new mesh so small that the lower bound decreases by at most Δ_j . Thus all finite chains using any of the levels have cost at least

$$\beta_* = \frac{1}{2} - \sum_j \Delta_j > \frac{1}{4}$$

from E to C_k . Each such chain uses only finitely many levels, so this assertion follows from the corresponding finite-stage bound.

For $x \in G$ let $D(x)$ be the infimum of the costs of these finite chains from E to x , allowing the value ∞ , and define

$$u^0(x) = \min\{1, 4D(x)\}. \tag{4.8}$$

The empty chain shows $u^0 = 0$ on E . A jump disk cannot cross C_k , because it is connected and its closure misses that curve. In a chain from E to a point of $U_k \cap G$, some true-path piece must therefore cross C_k . Truncating that chain at the crossing gives cost at least β_* . Consequently

$$u^0 = 0 \text{ on } E, \quad u^0 = 1 \text{ on } U_k \cap G. \tag{4.9}$$

Reversal and concatenation give, for every allowed true path from $x \in G$ to $y \in G$,

$$|u^0(x) - u^0(y)| \leq 4\mathcal{L}_{g_1, m}(\gamma). \tag{4.10}$$

This remains valid when one or both distances are infinite: a finite-cost connecting path makes finiteness of the two distances equivalent, and otherwise the inequality is vacuous or both truncated values are 1. Likewise any two domain points of a level- j jump disk V have value difference at most $4c_j(V)$.

Step 4: topological bands. For each $s \in B$, the level- j disk $V_j(s)$ containing s contains all domain points sufficiently close to s . Its oscillation bound therefore bounds the width of the topological band at s by $4c_j(V_j(s))$. We claim

$$\limsup_{j \rightarrow \infty} c_j(V_j(s)) \leq m(s).$$

Indeed, for every $\eta > 0$ the set $\{r \in B : m(r) \geq m(s) + \eta\}$ is finite and does not contain s . As the mesh and τ_j tend to zero, its points eventually lie outside the τ_j -neighborhood of $V_j(s)$. Also $\rho_j \rightarrow 0$. Equation (4.7) proves the claim. Hence every band has width at most $4m(s)$. In particular u^0 has a unique trace at every point of

$$B \setminus \Sigma', \quad \Sigma' = \Sigma \cup \{m > 0\}.$$

This is a countable enlargement of the flags.

Step 5: Sobolev estimates and band-preserving smoothing. The separating values and pointwise band bounds for u^0 are now established. It remains to obtain smoothness and the energy estimate.

Measurability and weak derivatives. The finite-chain codes retain all true-path tags and are Borel by Lemma 4.5: matching endpoints, anchor membership, and membership in specified jump disks are Borel conditions, and their costs are Borel. Projection over such codes shows that $\{x : D(x) < a\}$ is analytic for every a . Thus u^0 is measurable for completed area measure.

In a physical conformal chart, write $ds = \Lambda(z)|dz|$. Ordinary axis segments in the chart are allowed and contain no dust tolls. Equation (4.10) gives

$$|u^0(z_2) - u^0(z_1)| \leq 4 \int_{[z_1, z_2]} \Lambda g_1 |dz|$$

on every such segment. Fubini makes the line density integrable on almost every horizontal and vertical line in each compact chart rectangle. On each of these lines the displayed inequalities for all subsegments imply absolute continuity. Applying the usual slicing characterization of weak derivatives to an almost everywhere equal Borel representative gives

$$|\partial_x u^0|, |\partial_y u^0| \leq 4\Lambda g_1, \quad |du^0|_{\text{sph}} \leq 4\sqrt{2} g_1 \quad \text{almost everywhere.}$$

Since $0 \leq u^0 \leq 1$ and spherical area is finite, this is a global $W^{1,2}(G)$ function.

Smoothing with shrinking averaging neighborhoods. Choose a locally finite smooth partition of unity on relatively compact coordinate patches, and slightly larger patches in which to take positive mollified averages. The patches can be chosen so that every averaging sample contributing at a point x has X -distance from x tending to zero as x approaches B . For example, first require their X diameters to be at most a fixed small multiple of their distance to B , and then shrink the coordinate averaging radii further. On each patch approximate u^0 by a positive convolution average. Choose its value and derivative errors in L^2 so small that, including the derivatives of the partition functions, the sum of all errors is less than ε . Local Sobolev approximation permits this on each of the countably many relatively compact patches. The resulting smooth function u satisfies

$$0 \leq u \leq 1, \quad \|du - du^0\|_{2,G} < \varepsilon.$$

For precision, if ψ_j is the partition and v_j the local average, estimate

$$du - du^0 = \sum_j \psi_j (dv_j - du^0) + \sum_j (v_j - u^0) d\psi_j$$

by the triangle inequality in L^2 , assigning summable tolerances to both terms.

Preservation of bands and constants. To verify the band assertion pointwise at every dust point, let ℓ_s, r_s be the actual lower and upper topological limits of u^0 at $s \in B$. For every $\eta > 0$ all actual values of u^0 in some punctured domain neighborhood of s lie in $[\ell_s - \eta, r_s + \eta]$. Changing to an almost everywhere equal Borel representative preserves these bounds almost everywhere in that neighborhood. If $x \rightarrow s$ and a sample y contributes to the average at x , the patch construction gives, for a fixed constant c ,

$$d_X(y, s) \leq d_X(x, s) + c \operatorname{dist}_X(x, B) \rightarrow 0,$$

uniformly over the contributing patches and samples. Eventually all those averages, and then their positive partition sum, lie in the indicated interval. Letting $\eta \downarrow 0$ proves that all limiting values lie in the original band at s , without an exceptional dust set. Thus the smoothed bands are contained in those of u^0 , and every trace at a new free point is unchanged. Equation (4.9) also permits exact preservation of the required constants. Choose $l > k$ and arrange that every sampling patch meeting E_* lies in $\operatorname{int} E$, while every sampling patch meeting C_l lies in $U_k \cap G$. The two compact sets are disjoint and have the requisite open neighborhoods. Their averages are exactly 0 and 1, respectively.

The controlled size. Take $a_u = 4m$. Since $g_1 = g$ off O ,

$$\|du\|_{2,G \setminus O} + \|a_u\|_{\ell^2(B)} \leq 4\sqrt{2}\|g\|_2 + 4\|m\|_{\ell^2(B)} + \varepsilon \leq 8(\|g\|_2 + \|m\|_{\ell^2(B)}) + \varepsilon.$$

Total energy is finite as well, because $g_1 \in L^2(G)$.

Step 6: compatibility with every prior finite level union. Only (P) for the appended coordinate remains. We will show that its free trace values on each nondegenerate component of a generic prior finite level union form a null set.

Fix a type ι and choose a tuple outside N_ι and the null sets in Lemma 2.5 for the fixed weights g_1, m . Put $H = H_\iota(\mathbf{t})$. By Lemma 2.5 and its weighted coarea estimate, H has finite physical length in G , its nondegenerate components are locally connected and countable in number, and

$$\nu_H = \mathbf{1}_{H \cap G} g_1 d\mathcal{H}_{\text{sph}}^1 + \sum_{s \in H \cap B} m(s) \delta_s \quad (4.11)$$

is a finite Borel measure. The integrability behind this last assertion is, explicitly, bounded by the finite sum of

$$\int_{G \cap M_i} g_1 |du_i| da + \sum_{s \in B} m(s) a_i(s)$$

over the participating labels. Cauchy–Schwarz bounds each term. This also treats extended nonnegative values of g_1 , by truncation.

Let $\alpha \subset H$ be a simple arc joining two new free points $s, t \in B \setminus \Sigma'$. Its domain-part arclength is its physical image length and is finite. Give it the tag corresponding to the fixed permitted tuple. Subarcs between domain points are allowed, so Equation (4.10) bounds their value differences by four times their ν_H measure. Domain points occur arbitrarily close to both ends of α , since B is totally disconnected. Passing to the free trace limits gives

$$|u^0(s) - u^0(t)| \leq 4\nu_H(\alpha).$$

The same trace inequality holds for u , whose free traces were unchanged by smoothing. The measure ν_H gives zero mass to the new free dust points. Apply Lemma 2.6 in each nondegenerate component of H . The set of its free trace values has Lebesgue outer measure zero, hence zero image measure modulo 1. There are only countably many such components.

For the fixed generic prior tuple, almost every new level consequently contains no free point in a nondegenerate component of H . The universal measurability of the level-continuum incidence from Section 2 permits slicing in the tuple variables. This is exactly (P) for the appended coordinate. Older instances of (P) persist because enlarging the flags only removes free points. The barrier and all its asserted bounds are proved. $\square$

5 Positive laws of admissible paths

We now turn failure of a small compatible barrier into a probability law of paths with controlled mean arclength occupation and boundary contacts. The construction has two stages. Modulus duality supplies a law for each prescribed countable set of dust points, together with an auxiliary atomless measure on the dust. A fusion lemma then produces one law with square-summable hit probabilities on all of B .

Two requirements run through the argument. Admissibility of a metric must hold for every path, including paths that charge a reference-null set where the metric is infinite. The constants in the first stage must also be uniform in the prescribed countable set and atomless measure. These requirements allow the two stages to be combined.

Throughout this section the endpoint data of Definition 4.1 are fixed. The preceding coordinate list is countable and satisfies (P). The collection $\mathcal{Q}$ of preceding path laws is countable, and $m_\xi \in \ell^2(B)$ for every $\xi \in \mathcal{Q}$. Write

$$D_* = \Sigma \cup \bigcup_{\xi \in \mathcal{Q}} \{s \in B : m_\xi(s) > 0\}.$$

The set D_* is countable; it records the points at which the prior data will require control of visit multiplicity. A *base code* is a tagged path $\Gamma = (\gamma, \theta)$ satisfying Definition 4.2, including Equations (4.3) and (4.4), that starts in E , first reaches b at time 1, and respects the gates when gates are prescribed. We impose no auxiliary atomless-measure restriction or tuple exclusions on base codes. Their space, denoted by $\mathcal{A}$, is standard Borel by Lemma 4.5.

For a code $\Gamma = (\gamma, \theta)$, write $\ell_\Gamma := \nu_\gamma$ for the arclength occupation measure defined in Section 4, and put

$$N_\Gamma(s) = \#\{t \in [0, 1] : \gamma(t) = s\}, \quad N_\Gamma^-(s) = \#\{t \in [0, 1) : \gamma(t) = s\}.$$

An infinite cardinality in these expressions means $+\infty$. Thus $N_\Gamma = N_\Gamma^- + \mathbf{1}_{\{b\}}$, since the target is first reached at the final time. For a law μ on base codes, use the notation

$$\bar{\ell}_\mu = \int \ell_\Gamma d\mu(\Gamma), \quad n_\mu(s) = \mathbb{E}_\mu N_\Gamma(s), \quad m_\mu(s) = \mathbb{P}_\mu\{N_\Gamma(s) > 0\}.$$

Here n_μ counts visits with parameter multiplicity, whereas m_μ is a hit probability. In particular, $m_\mu \leq n_\mu$. We will retain the stronger visit-count estimate on D_* and obtain the hit-probability estimate on the entire dust.

Theorem 5.1 (Positive-law alternative). *Suppose that, for some $\beta > 0$, there is no new compatible barrier coordinate with*

$$\|du\|_{2,G \setminus O} + \|a_u\|_{\ell^2(B)} < \beta.$$

Nonexistence here allows an arbitrary countable enlargement of the flags and, in the gated problem, any penalty region permitted in Definition 4.1. Then there is a probability law μ on base codes with the following properties:

- (i) $\bar{\ell}_\mu = g_\mu \, da$, where $g_\mu \in L^2(G, da)$;
- (ii) $n_\mu|_{D_*} \in \ell^2(D_*)$ and $m_\mu \in \ell^2(B)$;
- (iii) for every slot and every tag type, its tuple distribution, regarded as a subprobability measure, is absolutely continuous with respect to Lebesgue measure on the corresponding tuple torus.

The three norms in (i)–(ii) are bounded by a constant depending only on β and the numerical constant in Theorem 4.3. In particular, the bound does not depend on any auxiliary countable subset of B or any auxiliary finite atomless measure on B .

To prove the theorem, we first establish the measure-family duality used in Lemma 5.3. That extraction lemma gives all the required estimates on prescribed countable data. We then prove Lemma 5.4 and check its hypotheses for the resulting family of path laws.

5.1 Duality for a Borel family of measures

Modulus for systems of measures goes back to Fuglede [6, Chapter I]. Its dual description by probability measures with controlled barycenter was developed by Ambrosio, Di Marino, and Savaré [2, Theorem 5.1 and Corollary 5.2(b)]. We give the $p = 2$ argument for the Borel families needed here. The input will be a positive lower bound for modulus; the output will be a law whose mean measure has a bounded L^2 density. The proof retains pointwise admissibility for extended densities and produces compact support after a refinement of the code topology.

If Y is Polish and σ is a finite Borel measure on Y , let $\mathcal{M}_+(Y)$ denote the space of finite positive Borel measures with the narrow topology. We write $\mathcal{P}(Y)$ for its subspace of probability measures. For a family $(\nu_a)_{a \in A}$ of such measures, define

$$\text{Mod}_\sigma(A) = \inf \left\{ \int_Y h^2 \, d\sigma : h : Y \longrightarrow [0, \infty] \text{ is Borel, } \int_Y h \, d\nu_a \geq 1 \text{ for every } a \in A \right\}.$$

The admissibility condition is pointwise in the index a ; no measures ν_a are discarded because of null sets for σ .

Lemma 5.2 (Measure-family duality). *Let Y be Polish, let σ be a finite Borel measure on Y , let A be standard Borel, and let $a \mapsto \nu_a \in \mathcal{M}_+(Y)$ be Borel. If $\text{Mod}_\sigma(A) > c > 0$, then, for every $C > c^{-1/2}$, there is a Borel probability measure P on A whose barycenter satisfies*

$$\int_A \nu_a \, dP(a) = q \, \sigma, \quad q \geq 0, \quad \|q\|_{L^2(\sigma)} \leq C.$$

Moreover, after giving A a Polish topology with its original Borel sets and with $a \mapsto \nu_a$ continuous, the law may be supported on a compact subset of A .

Proof. The Polish refinement theorem gives the topology in the statement; see [10, Section 13.A, in particular Exercise 13.5]. We work in this topology. First we prove continuity under increasing unions and approximation by open neighborhoods, keeping the prescribed values of every density on reference-null sets. These two properties will give a compact subfamily of positive modulus. Separation on that compact subfamily will then produce the required probability law.

Increasing families. If $A_j \uparrow A_\infty$, then

$$\text{Mod}_\sigma(A_\infty) = \lim_j \text{Mod}_\sigma(A_j). \quad (5.1)$$

Only the case where the limit M is finite needs proof. Choose nonnegative Borel admissible metrics h_j for A_j whose squared norms tend to M . Weak compactness in $L^2(\sigma)$, followed by convex combinations of tails, gives a nonnegative limit h and finite convex combinations v_k of metrics with indices at least k , such that

$$\|h\|_2 \leq \sqrt{M}, \quad \sum_k \|v_k - h\|_2 < \infty.$$

Choose a Borel representative of h finite everywhere. Keep the actual extended Borel representatives of the v_k , and set $D = \sum_k |v_k - h|$. This is an extended nonnegative Borel function with $\|D\|_2 < \infty$. For $a \in A_\infty$, the metrics v_k are admissible at a for all sufficiently large k . If $\int D d\nu_a = \infty$, then $h + \varepsilon D$ is admissible there for every $\varepsilon > 0$. Otherwise $\int |v_k - h| d\nu_a \rightarrow 0$, so either $\int h d\nu_a = \infty$ or convergence of the integrals gives $\int h d\nu_a \geq 1$. Consequently $h + \varepsilon D$ is admissible on the entire union. Letting $\varepsilon \downarrow 0$ proves Equation (5.1). The added density D accounts for every index at which convergence in $L^2(\sigma)$ alone would not control integration against ν_a .

Open neighborhoods. For every subset $F \subset A$ and every $q > \text{Mod}_\sigma(F)$, there is an open set $V \supset F$ with $\text{Mod}_\sigma(V) < q$. To see this, first note that every nonnegative Borel $h \in L^2(\sigma)$, including its prescribed extended values on null sets, has a nonnegative lower semicontinuous pointwise majorant with arbitrarily little increase of its L^2 norm. Indeed, for $e > 0$ set

$$h_e = e \sum_{j=0}^{\infty} \mathbf{1}_{\{h > je\}}.$$

At finite values, $h \leq h_e \leq h + e$. By outer regularity, choose open sets $U_j \supset \{h > je\}$ for which $e \sum_j \sigma(U_j \setminus \{h > je\})^{1/2}$ is arbitrarily small. Then $H = e \sum_j \mathbf{1}_{U_j}$ is lower semicontinuous, majorizes h pointwise, and satisfies

$$\|H\|_2 \leq \|h\|_2 + e\sigma(Y)^{1/2} + e \sum_j \sigma(U_j \setminus \{h > je\})^{1/2}.$$

Apply this construction to an admissible metric for F of squared norm less than q , and scale its majorant slightly so that its integrals over F are strictly greater than 1, still with squared norm less than q . The set $V = \{a : \int H d\nu_a > 1\}$ is open by lower semicontinuity of integration and continuity of the measure map. It has the required modulus bound.

A compact subfamily. We now use the two modulus properties to retain positive modulus while restricting the codes to a compact set. Use a complete compatible metric on A . At step j , cover A by countably many closed balls of radius 2^{-j} . Starting with $F_0 = A$, Equation (5.1) permits selection of a finite union of these balls whose intersection F_j with F_{j-1} still has modulus greater than c . The sets F_j are nested and closed. Every sequence $a_j \in F_j$ has a Cauchy subsequence, by the finite ball covers at each fixed depth. Completeness shows that $K = \bigcap_j F_j$ is nonempty and compact. The same argument shows that every open neighborhood of K contains all sufficiently late F_j . If $\text{Mod}_\sigma(K) < c$, the open-neighborhood property would therefore contradict $\text{Mod}_\sigma(F_j) > c$. Hence $\text{Mod}_\sigma(K) \geq c$.

The compactness here comes from the nested finite covers and completeness. Individual closed balls in the refined topology need not be compact.

Separation and the barycenter. For every bounded continuous $h \geq 0$ with $\|h\|_2 \leq 1$, some $a \in K$ satisfies $\int h d\nu_a \leq C$. Otherwise h/C would be admissible on K , contrary to $C^{-2} < c \leq \text{Mod}_\sigma(K)$. Given finitely many such tests $h_1, \dots, h_n$, consider their integral vectors averaged against probability measures on K . This is a compact convex subset of $\mathbb{R}^n$. If it missed the orthant $(-\infty, C]^n$, strict separation would give $\alpha_j \geq 0$, $\sum_j \alpha_j = 1$, such that $\sum_j \alpha_j \int h_j d\nu_a > C$ for every $a \in K$. The test $\sum_j \alpha_j h_j$ would contradict the preceding observation. Thus a probability measure meets all the finitely many inequalities.

Compactness of the probability measures on K , and continuity of the integral tests there, now give one law P meeting every bounded continuous nonnegative test. Its barycenter η is finite: the continuous mass function $a \mapsto \nu_a(Y)$ is bounded on K . Scaling the tests, including arbitrarily large multiples of tests of zero L^2 norm, gives

$$\int h d\eta \leq C \|h\|_{L^2(\sigma)} \quad (h \geq 0 \text{ bounded and continuous}).$$

Apply this also to $|h|$. Density of bounded continuous functions in $L^2(\sigma)$ and the Hilbert-space representation theorem give a nonnegative density q of norm at most C . The finite measures η and $q\sigma$ agree on bounded continuous functions and hence are equal. $\square$

5.2 Countable and atomless tests

We apply the duality lemma to three measures associated with a tagged path: its domain arclength occupation, its visits to a prescribed countable set, and a weighted measure of its tag tuples. The first two will give the occupation and visit-count estimates. The third will make every slot's tuple law absolutely continuous. Together with the residual atomless-measure constraint, this will control the whole path image under the prescribed atomless measure.

Let $\mathcal{I}$ be the countable set of tag types $\iota = (i_1, \dots, i_d)$, and write $d(\iota) = d$. Fix positive numbers ω_ι with sum at most 1. Use the Polish disjoint union and finite reference measure

$$\Pi = \coprod_{\iota \in \mathcal{I}} \mathbb{T}^{d(\iota)}, \quad \tau = \sum_{\iota \in \mathcal{I}} \omega_\iota d\mathbf{t}.$$

If there are no tag types, Π is empty. For a code Γ , define its weighted tuple measure by

$$\psi_\Gamma = \sum_{k \text{ used}} 2^{-k} \delta_{(\iota_k, \mathbf{t}_k)}. \quad (5.2)$$

The components of Π are indexed by *types*, whereas the factors 2^{-k} are indexed by optional *slots*. This distinction allows tags to be reassigned to later slots without changing their geometric data.

Lemma 5.3 (Prescribed-countable-set extraction). *Under the hypothesis of Theorem 5.1, there is a constant $C < \infty$, depending only on β and the constant in Theorem 4.3, with the following property. For every countable $D \supset D_* \cup \{b\}$ and every finite atomless Borel measure λ on B , there is a law μ on base codes such that*

$$\bar{\ell}_\mu = g_\mu da, \quad \|g_\mu\|_2 \leq C, \quad (5.3)$$

$$\|n_\mu|_D\|_{\ell^2(D)} \leq C + 1, \quad \int_B m_\mu d\lambda = 0, \quad (5.4)$$

$$\mathbb{E}_\mu \psi_\Gamma = q_\mu \tau, \quad \|q_\mu\|_{L^2(\tau)} \leq C. \quad (5.5)$$

In addition, $\lambda(\gamma(Z_\Gamma)) = 0$ for almost every code.

Proof. The reference space and path measures. Choose $c_s > 0$, $s \in D$, with $\sum_{s \in D} c_s^2 \leq 1$, and put

$$Y = G \sqcup D_{\text{disc}} \sqcup \Pi, \quad \sigma = da \oplus \sum_{s \in D} c_s^2 \delta_s \oplus \tau.$$

Add the constraint $\lambda(\gamma(Z_\Gamma)) = 0$ to the base path system. For its codes define

$$\nu_\Gamma = \ell_\Gamma \oplus \sum_{s \in D} c_s N_\Gamma^-(s) \delta_s \oplus \psi_\Gamma. \quad (5.6)$$

Restrict to codes for which this measure is finite. This is a standard Borel family, and its map to $\mathcal{M}_+(Y)$ is Borel by Lemma 4.5. In particular, the mean measures used below are ordinary Borel measure kernels.

A modulus bound independent of the prescribed tests. Let C_{bar} denote the numerical constant in Theorem 4.3. The modulus of this family is at least

$$c_0 = \left(\frac{\beta}{4C_{\text{bar}}} \right)^2. \quad (5.7)$$

Suppose, to the contrary, that it has an admissible extended Borel metric h with $\|h\|_{L^2(\sigma)} < \beta/(4C_{\text{bar}})$. Its domain and atom parts give

$$g_0 = h|_G, \quad m_0(s) = \begin{cases} c_s h(s), & s \in D, \\ 0, & s \notin D, \end{cases} \quad \|g_0\|_2^2 + \|m_0\|_{\ell^2(B)}^2 \leq \|h\|_{L^2(\sigma)}^2.$$

For a sufficiently small $\delta > 0$, set $g = g_0 + \delta$ and $m = m_0 + \delta \sum_{s \in D} c_s \mathbf{1}_{\{s\}}$. Then g is bounded below by a positive constant and

$$C_{\text{bar}}(\|g\|_2 + \|m\|_{\ell^2(B)}) < \beta/2.$$

The small additions are permitted because $\text{area}(G) < \infty$ and $\|(c_s)\|_{\ell^2(D)} \leq 1$.

In each type exclude the set on which the corresponding tuple part of h is infinite. It is Lebesgue null, because that component has a positive reference weight. We claim that every allowed path for this additional null-exclusion system has true (g, m) -cost at least 1, omitting the final target toll. It suffices to consider one of finite true cost. The added tolls imply $\sum_{s \in D} c_s N_\Gamma^-(s) < \infty$, so the non-tuple parts of Equation (5.6) are finite. All the tagged tuples have finite h -value. Reassign the countably many tags to successively larger unused slots so that the integral of the tuple part of h is less than any prescribed $\varepsilon > 0$. This leaves the path, tags, residual set, and every path-system constraint unchanged. The resulting code belongs to the family on which h is admissible. Since its non-tuple integrals are bounded by the true cost,

$$1 \leq \int h d\nu_\Gamma \leq \mathcal{L}_{g,m}^-(\gamma) + \varepsilon.$$

Here the superscript minus means that only the final target time is omitted. Let $\varepsilon \downarrow 0$. Paths of infinite cost satisfy the same lower bound automatically.

Theorem 4.3 applies to this path system, with its atomless constraint and its additional null tuple exclusions. Taking the error in that theorem sufficiently small produces a compatible barrier of budget less than β , a contradiction. This proves Equation (5.7). Notice that its constant does not depend on D , λ , or the choices of c_s .

Occupation, visit counts, and tuple distributions. Apply Lemma 5.2 with $c = c_0/2$ and any fixed $C > \sqrt{2/c_0}$. The resulting law has mean measure with density of $L^2(\sigma)$ norm at most C . On G and Π , this gives Equations (5.3) and (5.5). At $s \in D$, the mean mass is $c_s \mathbb{E} N_\Gamma^-(s)$, while the reference mass is c_s^2 . Consequently the contribution of the atom part to squared density norm is exactly

$$\sum_{s \in D} (\mathbb{E} N_\Gamma^-(s))^2.$$

Restoring the final visit adds one coordinate vector of norm 1, proving the count estimate in Equation (5.4).

The atomless test on the whole path image. It remains to verify the atomless-measure conclusion, which concerns the whole path image. Equation (5.5) implies absolute continuity of the tuple distribution in every fixed slot and type: that distribution, multiplied by 2^{-k} , is dominated by the mean weighted tuple measure. At a fixed free point s , incidence of a tag of type $(i_1, \dots, i_d)$ requires at least one of its levels to equal the unique trace of the corresponding coordinate at s , modulo 1. This is a finite union of Lebesgue-null coordinate sections of the tuple torus. Thus a fixed free point is visited during an open tag interval with probability zero. Since $\lambda(\Sigma) = 0$, Fubini and the countable number of slots and types show that the expected λ -measure of all tagged dust contacts is zero. The residual contacts have λ -measure zero on every code in the family. Combining the two parts gives $\int_B m_\mu d\lambda = 0$. $\square$

5.3 Fusion from countable tests to the whole dust

The preceding lemma permits the law to depend on the countable set D and the atomless measure λ . We now need a single law whose hit probabilities satisfy a common bound on every finite subset of B . The next lemma makes this passage using joint upper semicontinuity and closure under Borel probability mixtures. Its proof constructs mixtures with estimates that hold throughout prescribed neighborhoods, then keeps all later choices inside those neighborhoods.

Lemma 5.4 (Fusion of laws). *Let S be standard Borel, let B be compact metric, and let $\mathcal{M}$ be a Borel family of probability measures on S , closed under Borel probability mixtures. Suppose a Borel relation $\mathcal{H} \subset S \times B$ defines*

$$m_z(s) = z\{x \in S : (x, s) \in \mathcal{H}\} \in [0, 1].$$

Assume that $\mathcal{M}$ has a Polish topology generating its Borel sets for which $(z, s) \mapsto m_z(s)$ is jointly upper semicontinuous. Finally, suppose that for every countable $D \subset B$ and every finite atomless Borel measure λ on B , some $z \in \mathcal{M}$ satisfies

$$\|m_z|_D\|_{\ell^2(D)} \leq C_0, \quad \int_B m_z d\lambda = 0. \quad (5.8)$$

Then, for every $\varepsilon > 0$, there exists $\bar{z} \in \mathcal{M}$ with $\|m_{\bar{z}}\|_{\ell^2(B)} \leq C_0 + \varepsilon$.

Proof. Fix a complete compatible metric on $\mathcal{M}$. The defining relation and Tonelli's theorem give the mixture identity

$$m_{\int z dP(z)}(s) = \int m_z(s) dP(z) \quad (s \in B) \quad (5.9)$$

for every Borel probability measure P on $\mathcal{M}$.

The small-set ideal. We first identify the subsets of candidate laws within which every prescribed countable and atomless test can still be met. Call $A \subset \mathcal{M}$ small if some countable D and finite

atomless λ admit no member of A satisfying Equation (5.8). Small sets form a sigma-ideal. Indeed, for a countable union take the union of the witnessing countable sets and a strictly positively weighted sum of the witnessing measures, with weights chosen to give finite total mass. A law satisfying the two tests for these combined data would satisfy the tests for each summand. By hypothesis $\mathcal{M}$ is not small.

For a nonsmall closed set A , remove every basis-open set V for which $A \cap V$ is small. The remaining set A' is closed and nonsmall. Every neighborhood of each point of A' meets A nonsmally. Only countably many basis-open sets are removed, which is the reason this conclusion follows from the sigma-ideal property.

A finite split. Fix a closed nonsmall set A , let A' be the set just obtained, and fix $r > 0$. An r -separated tuple in B has bounded cardinality, say at most N_r . Consider the compact family T_r of measures

$$\sum_{j=1}^n w_j \delta_{s_j}, \quad 0 \leq n \leq N_r, \quad \text{dist}(s_i, s_j) \geq r \ (i \neq j), \quad w_j \geq 0, \quad \sum_j w_j^2 \leq 1.$$

The empty tuple gives the zero measure. Its probability barycenters form a compact convex set K_r of positive measures on B . For every $\zeta \in K_r$, the weights of its atomic part have ℓ^2 norm at most 1: on any fixed finite point set this follows from convexity of the Euclidean norm, and one then takes the supremum over finite sets. Write $\zeta = \zeta_{\text{at}} + \lambda$, where λ is atomless. Nonsmallness of A' , tested against the countable atoms of ζ and this λ , gives a point $z \in A'$ with

$$\int m_z d\zeta = \int m_z d\zeta_{\text{at}} \leq C_0.$$

The following finite-mixture step uses the classical minimax strategy of the Kneser–Fan theorem as presented by Sion [20, Theorem 4.2, p. 175]. We give the finite-cover and separation proof here; the surrounding fusion construction is developed locally in this proof.

For fixed z , the function $\zeta \mapsto \int m_z d\zeta$ is affine and upper semicontinuous on K_r . Therefore the open sets

$$\left\{ \zeta \in K_r : \int m_z d\zeta < C_0 + \varepsilon/4 \right\}, \quad z \in A',$$

cover K_r . Choose a finite subcover with centers $z_1, \dots, z_k$. Their integral-vector image of K_r is convex and misses the open orthant in which every coordinate exceeds $C_0 + \varepsilon/4$. Separation from this open convex set gives coefficients $\alpha_j \geq 0$, $\sum_j \alpha_j = 1$, such that

$$\sup_{\zeta \in K_r} \sum_j \alpha_j \int m_{z_j} d\zeta \leq C_0 + \varepsilon/4.$$

No closedness of the integral-vector image is needed here. The separator has nonnegative coefficients because the orthant is unbounded in every positive coordinate direction. Testing on T_r and using nonnegative Euclidean dual weights yields

$$\sup_{F \text{ } r\text{-separated}} \left\| \sum_j \alpha_j m_{z_j} \right\|_F \leq C_0 + \varepsilon/4. \quad (5.10)$$

To extend this estimate to neighborhoods of the centers, keep the coefficients α_j fixed and

write its left side as

$$\begin{aligned}\Phi_r(y_1, \dots, y_k) &= \sup_{F \text{ } r\text{-separated}} \left\| \sum_{j=1}^k \alpha_j m_{y_j}|_F \right\|_{\ell^2(F)} \\ &= \max_{\substack{0 \leq n \leq N_r, \ s_1, \dots, s_n \in B \\ \text{dist}(s_i, s_l) \geq r \ (i \neq l) \\ w_i \geq 0, \ \sum_{i=1}^n w_i^2 \leq 1}} \sum_{i=1}^n \sum_{j=1}^k w_i \alpha_j m_{y_j}(s_i).\end{aligned}$$

The case $n = 0$ contributes zero. The configuration and weight spaces in this maximum form a finite union of compact spaces. Joint upper semicontinuity of the hit functions therefore makes Φ_r upper semicontinuous jointly in its centers. The margin in Equation (5.10) now gives each center a small closed neighborhood V_j , of any prescribed sufficiently small diameter, such that

$$\left\| \sum_j \alpha_j m_{y_j}|_F \right\|_{\ell^2(F)} < C_0 + \varepsilon \quad (5.11)$$

for every simultaneous choice $y_j \in V_j$ and every r -separated finite set F . Take the neighborhoods to be closed balls contained in the requisite open neighborhoods. Each child $A \cap V_j$ is closed and nonempty: the ball V_j contains an open neighborhood of its center, and every such neighborhood meets A nonempty. Discard children with zero coefficient.

Nested splits. Starting with $A = \mathcal{M}$, perform the split at successive depths with $r_n \downarrow 0$. At depth n , require every child to have diameter at most 2^{-n} . The result is a finitely branching tree of nonempty nested closed sets, with transition probabilities given by the coefficients α_j . Completeness gives a unique point of $\mathcal{M}$ along every infinite branch. The branch-limit map is continuous: two branches with the same depth- n ancestor have limits in a set of diameter at most 2^{-n} . The transition probabilities give a Borel probability measure on branches and thus a Borel probability measure P on their limits. By mixture closure, $\bar{z} = \int z \, dP(z)$ belongs to $\mathcal{M}$.

It remains to check every finite-set estimate for this one law $\bar{z}$. Fix a depth n and an r_n -separated finite set F . At a parent of that depth, let P_j be the distribution of the branch limit conditioned on taking its j th child. Its transition probability is α_j , and P_j is supported in that child's closed set. These conditionings refer to tree nodes, so the closed child sets are allowed to overlap. Equation (5.11) holds for every simultaneous choice of points from the child sets. Integrating with respect to the product of their conditional distributions gives

$$\begin{aligned}\left\| \sum_j \alpha_j \int m_{y_j}|_F \, dP_j(y) \right\|_{\ell^2(F)} &\leq \int \left\| \sum_j \alpha_j m_{y_j}|_F \right\|_{\ell^2(F)} \, d(P_1 \otimes \dots \otimes P_k)(y_1, \dots, y_k) \\ &\leq C_0 + \varepsilon.\end{aligned}$$

The first inequality is convexity of the Euclidean norm. Its left side is the norm of the conditional mean hit vector at the parent. Average over the finitely many parents at this depth, and use Equation (5.9), to obtain $\|m_{\bar{z}}|_F\|_2 \leq C_0 + \varepsilon$. Every finite set of distinct points is r_n -separated for all sufficiently large n . Taking the supremum over finite sets proves the lemma. $\square$

5.4 Application of fusion to path laws

We apply fusion to the family of laws retaining the occupation, visit-count, and tuple-density bounds from Lemma 5.3.

Proof of Theorem 5.1. Fix the reference measure τ on types and the constant C from Lemma 5.3. Let $\mathcal{M}$ be the laws z on the base code space $\mathcal{A}$ satisfying

$$\bar{\ell}_z = g_z \, da, \quad \|g_z\|_2 \leq C; \quad \|n_z|_{D_*}\|_{\ell^2(D_*)} \leq C + 1; \quad \mathbb{E}_z \psi_\Gamma = q_z \tau, \quad \|q_z\|_2 \leq C. \quad (5.12)$$

We check the structural hypotheses of Lemma 5.4.

Borel structure and mixture closure. For a Borel kernel of finite measures on a Polish space with finite reference measure σ_0 , the condition that its mean have nonnegative $L^2(\sigma_0)$ density of norm at most C is Borel. To verify this directly, choose a countable algebra generating the Borel sets, and test

$$\int \left(\int h \, d\nu_\Gamma \right) dz(\Gamma) \leq C \|h\|_{L^2(\sigma_0)}$$

for all nonnegative rational simple functions on that algebra, including the constant function. These are countably many Borel inequalities. The constant test makes the mean measure finite. The inequalities define a bounded functional on the dense space of algebra-simple functions in $L^2(\sigma_0)$; applying the estimate to absolute values treats signed functions. Its representing density gives the mean measure, first on the algebra and then on all Borel sets. Thus the tests are also sufficient.

Apply this observation to ℓ_Γ and ψ_Γ . The count functions $N_\Gamma(s)$ are Borel by Lemma 4.5; finite partial sums of the squared mean counts test the middle condition of Equation (5.12). It follows that $\mathcal{M}$ is a Borel subset of the standard Borel space of probabilities on $\mathcal{A}$. The same integral inequalities and convexity of finite-dimensional Euclidean norms show that all three conditions are preserved by every Borel probability mixture of members of $\mathcal{M}$. A mixture is still a law on the base code space, so it also preserves the endpoint conditions and path-system constraints.

Upper semicontinuity of hits. Forgetting tags gives a Borel map

$$\mathcal{M} \longrightarrow \mathcal{P}(C([0, 1], X)).$$

Choose a Polish topology on $\mathcal{M}$, with its same Borel sets, making this map continuous. The Borel probability mixtures just checked remain the same under this refinement. The relation

$$\mathcal{H}_0 = \{(\gamma, s) \in C([0, 1], X) \times B : \gamma(t) = s \text{ for some } t \in [0, 1]\}$$

is closed: a sequence of witnessing times has a convergent subsequence. If $z_j \rightarrow z$ in the chosen topology and $s_j \rightarrow s$, the products of their untagged path laws with δ_{s_j} converge weakly to the corresponding product for z, s . The closed-set inequality therefore gives

$$\limsup_j m_{z_j}(s_j) \leq m_z(s).$$

The hit functions have the joint upper semicontinuity required by Lemma 5.4. Their defining relation on tagged codes is Borel by the same closed hit test composed with the forgetful map.

The countable tests. The structural hypotheses are now established. The uniform extraction lemma supplies the remaining test hypothesis within this fixed candidate family. Given any countable $D \subset B$ and finite atomless λ , apply Lemma 5.3 to $D \cup D_* \cup \{b\}$. The resulting law belongs to $\mathcal{M}$, and $m_z \leq n_z$ gives

$$\|m_z|_D\|_{\ell^2(D)} \leq C + 1, \quad \int_B m_z \, d\lambda = 0.$$

All hypotheses of Lemma 5.4 hold with $C_0 = C + 1$. Take $\varepsilon = 1$ there. Its law $\mu \in \mathcal{M}$ satisfies $\|m_\mu\|_{\ell^2(B)} \leq C + 2$, while membership in $\mathcal{M}$ retains the occupation and mean-count estimates. Finally the last condition in Equation (5.12) implies absolute continuity in each slot and type, by the positive domination in Equation (5.2). This proves every conclusion with the common bound $C + 2$. $\square$

6 The first selection and path-level intersections

This section has two outputs. First, we select a countable compatible list of real barriers and a countable set of ends S . In every later compatible finite-transfer limit, each component indexed by $B \setminus S$ will be a point. The selection rests on a planar crossing argument: uncountably many endpoints without small barriers would give laws whose forced intersections contradict their occupation and hit bounds. Second, we prove that a path law has finite expected intersection count with a generic prior level union and avoids its entire free dust subset. This will supply the intersection estimates for the period coordinates.

We retain g_μ for a law's mean domain arclength density, n_μ for its expected visit counts including the final time, and m_μ for its hit probabilities. Thus $m_\mu \leq n_\mu$. Theorem 5.1 bounds m_μ on all of B and n_μ on the prescribed countable set D_* ; the distinction will matter in the intersection estimates.

6.1 Point avoidance and intersections of two laws

We begin by ruling out atoms at physical domain points. The corresponding statement for dust points requires the residual incidence constraints and the absolutely continuous tag distributions.

Lemma 6.1 (Avoidance of domain points). *Let μ be a law of paths ending in B , with expected domain arclength occupation $g_\mu da$ and $g_\mu \in L^2(G)$. Every fixed point of G is visited with probability zero. In particular, if all paths start outside the closure of a dust disk U and end in $U \cap B$, their last-hit locations on ∂U have an atomless distribution.*

Proof. Fix $x \in G$. In a physical coordinate disk about x , with coordinate $z(x) = 0$, choose $r_0 > 0$ so that its closed r_0 -disk lies in G . Consider the nonnegative density, defined to be zero at x and outside the indicated punctured disk,

$$\rho(z) = \frac{\mathbf{1}_{\{0 < |z| < r_0/e\}}}{|z| \log(r_0/|z|)}.$$

The smooth spherical length factor is bounded above and below on this disk, and

$$\int_0^{r_0/e} \frac{dr}{r \log^2(r_0/r)} < \infty, \quad \int_0^{r_0/e} \frac{dr}{r \log(r_0/r)} = \infty.$$

Consequently $\rho \in L^2(G, da)$, while every path that visits x and subsequently leaves this coordinate disk has infinite ρ -length on its domain times. To see the latter assertion without any regularity at the hitting time, stop between the circles of radii a and r_0/e and apply ordinary variation to the radial coordinate, then let $a \downarrow 0$. Every path under consideration ends in B and therefore must leave this disk after a visit to x . On the other hand,

$$\mathbb{E}_\mu \int_{\gamma^{-1}(G)} \rho(\gamma) ds = \int_G \rho g_\mu da \leq \|\rho\|_2 \|g_\mu\|_2 < \infty.$$

This proves point avoidance.

For a path from outside $\bar{U}$ to $U \cap B$, the set of hitting times of ∂U is a nonempty compact set whose maximum is less than 1. The path lies strictly inside U after that maximum. An atom at a given last-hit location would give positive probability of visiting that fixed point of $\partial U \subset G$, which has just been excluded. $\square$

Lemma 6.2 (Removal of diffuse dust intersections). *Fix a countable prior coordinate list satisfying (P), with flag set Σ . Let ξ and μ be laws on allowed tagged paths for this list, and put $A_\xi = \{s \in B : m_\xi(s) > 0\}$. Assume that*

$$m_\xi, m_\mu \in \ell^2(B), \quad n_\xi|_\Sigma \in \ell^2(\Sigma), \quad n_\mu|_{A_\xi} \in \ell^2(A_\xi).$$

Assume also that, for each law and every slot and tag type, the corresponding tuple subprobability is absolutely continuous with respect to Haar measure on its tuple torus. Finally, assume that ξ is imposed as an earlier law in the path system for μ . For independent samples from ξ and μ , almost surely,

$$\gamma_\xi([0, 1]) \cap \gamma_\mu([0, 1]) \cap B \subset \Sigma \cup \{m_\xi > 0\} \cup \{m_\mu > 0\}.$$

In particular all common dust visits are confined, up to a null event, to a countable set.

These hypotheses hold for the laws used below. Indeed, Theorem 5.1 supplies the hit and tuple bounds; the prescribed countable set for ξ contains Σ , and that for μ contains the hit support of each imposed earlier law.

Proof. There are three cases to control: a residual contact of the later path, a residual contact of the earlier path with a later tag, and a contact between two tags.

Residual contacts of the later path. Let P_N be the finite deletions exhausting A_ξ in the residual intersection constraint. With Z_γ denoting the residual dust times of a tagged path, that constraint, independence, and Tonelli's theorem give

$$\begin{aligned} & \mathbb{P}(\gamma_\xi([0, 1]) \cap (\gamma_\mu(Z_{\gamma_\mu}) \setminus P_N) \neq \emptyset) \\ & \leq \sum_{s \in A_\xi \setminus P_N} m_\xi(s) n_\mu(s) \leq \|m_\xi \mathbf{1}_{A_\xi \setminus P_N}\|_{\ell^2} \|n_\mu|_{A_\xi}\|_{\ell^2}. \end{aligned}$$

The last factor is finite by the assumed count bound, and the first tends to zero. Thus any meeting at a residual time of the later path occurs in A_ξ , almost surely.

An earlier residual contact with a later tag. Next fix a prior coordinate i , and delete finite sets exhausting Σ in its residual level constraint. Almost surely the weighted sum of visits of γ_ξ against a_i is finite, since its expectation is at most

$$\|a_i\|_{\ell^2} \|n_\xi|_\Sigma\|_{\ell^2}.$$

The deleted tails of this sum tend to zero. It follows directly from the residual level constraint that

$$|\{t \in \mathbb{T} : H_i(t) \cap (\gamma_\xi(Z_{\gamma_\xi}) \setminus \Sigma) \neq \emptyset\}| = 0 \quad \text{almost surely.}$$

This assertion holds simultaneously for all prior labels. Conditional on an earlier path with this property, a tag of the independent later path cannot meet its residual image off Σ : each coordinate level in the tag has an absolutely continuous marginal distribution. There are only countably many tag slots and types, so all such meetings are excluded simultaneously.

Contacts between two tags. It remains to exclude a free common point lying on a tag of each path. For any fixed pair of slots and types, their two tuples are independent and absolutely continuous with respect to their level product measures. We can therefore assume simultaneously that all required instances of (P) hold and that levels belonging to the same label in the two tuples are distinct. At a free common point x , choose the largest label j among the levels from either tag

that actually contain x . The unique trace at x implies that label j can occur at x in only one of the two tags. In the other tag, every level present at x has label less than j .

Here the other tag supplies a nondegenerate continuum through x in those lower levels. Indeed its image is connected and nonconstant. Choose a compact parameter subinterval within the tag whose image contains both x and another point. Level sets in its finite union that miss x miss some neighborhood of x , by closedness. Boundary bumping in a sufficiently small closed neighborhood now gives a nondegenerate subcontinuum through x using only the levels present there. This contradicts (P) for label j .

All these assertions hold simultaneously for the countably many pairs of slots, types, and sublists. The incidence events are Borel or analytic by the path coding and compact level graphs; completed product measures therefore justify the conditioning and Fubini arguments. $\square$

6.2 The crossing argument

We now compare two consequences of having many unavailable endpoints. Four alternating collar starts force an intersection, while localization of the law bounds makes each possible cross-pair intersection unlikely.

Theorem 6.3 (Countably many unavailable endpoints). *Fix any countable coordinate list satisfying (P), and fix $\beta > 0$. In the ungated endpoint problem with*

$$E = \{3R \leq |z| \leq 4R\}, \quad E_* = \{|z| = 7R/2\},$$

there are at most countably many $b \in B$ for which no compatible real barrier u is available with

$$\|du\|_2 + \|a_u\|_{\ell^2(B)} < \beta.$$

Availability permits a countable enlargement of the flags. The prescribed collar sequence may start arbitrarily deep, and the barrier is required to equal one on some collar of that sequence.

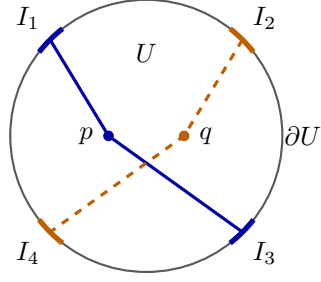
Proof. A common collar for uncountably many laws. Suppose the unavailable endpoints form an uncountable set. Choose distinct such endpoints b_α , $\alpha < \omega_1$. Recursively apply Theorem 5.1 to obtain a law μ_α from E to b_α at first arrival. At stage α , impose all the earlier laws in the residual intersection constraint. This is an admissible countable collection because α is countable. The coordinate list and its flags remain fixed throughout this recursion.

Fix a number $\delta > 0$, to be made small below. For each endpoint $p = b_\alpha$, shrinking its exhaustion disk gives

$$\|g_{\mu_p}\|_{2,U \cap G} < \delta, \quad \|m_{\mu_p}|_{(U \cap B) \setminus \{p\}}\|_{\ell^2} < \delta \tag{6.1}$$

for some exhaustion disk U containing p . The first assertion is absolute continuity of the energy integral, since the disks shrink to a point outside G . The second is the analogous tail property of a square-summable family after omitting its value at p . There are only countably many disks in the exhaustion. Hence one common U satisfies Equation (6.1) for uncountably many of the chosen endpoints and their laws.

Alternating connections. For each of these laws, take the last-hit location on ∂U . Its distribution is atomless by Lemma 6.1. Consequently there are four pairwise disjoint arcs I_1, I_2, I_3, I_4 on ∂U , listed in circular order, each having probability at least $1/10$. The arcs can be chosen from a fixed countable family: use any fixed parametrization of the circle and endpoints in a countable dense set, choosing slightly smaller quantile arcs before this approximation. After another uncountable



Two tails to p connect I_1 to I_3 .
Two tails to q connect I_2 to I_4 .
The alternating connections must meet in U .

Figure 2: The four-path crossing in the quotient disk U . The four boundary arcs occur in cyclic order. After diffuse dust contacts have been removed, the probability of any required meeting is bounded by the occupation and hit estimates.

restriction the same ordered four arcs work for every law. Choose an earlier endpoint p and a later endpoint q from this subfamily.

Independently sample two paths to p and two paths to q . With probability at least 10^{-4} their last-hit locations fall, respectively, in I_1 and I_3 for the p -paths and I_2 and I_4 for the q -paths. The tails after these hits lie strictly inside U , apart from their initial points. Concatenating the two tails through p gives a connection between its two boundary locations, and likewise for q . Each connection contains a simple arc between its boundary endpoints. Jordan separation in the closed disk $\bar{U}$ forces the two connections to intersect in U ; see Figure 2. Thus this event entails an intersection in U of one of the four independent cross-pairs consisting of a p -path and a q -path.

Dust intersections. We estimate the probability of such a cross-pair intersection. Lemma 6.2 removes all dust meetings outside a countable set. Points in that set with zero hit probability for either law can also be discarded. Independence and the union bound therefore give

$$\begin{aligned} \mathbb{P}(\text{a common visit in } U \cap B) &\leq \sum_{s \in U \cap B} m_{\mu_p}(s) m_{\mu_q}(s) \\ &\leq m_{\mu_q}(p) + m_{\mu_p}(q) + \sum_{s \in (U \cap B) \setminus \{p, q\}} m_{\mu_p}(s) m_{\mu_q}(s) \\ &\leq 2\delta + \delta^2. \end{aligned} \tag{6.2}$$

Here the target hit probabilities equal one, $p \neq q$, and the last sum is estimated by Cauchy–Schwarz using Equation (6.1).

Domain intersections. For domain meetings, let ℓ_γ denote a sample’s arclength occupation restricted to $U \cap G$, and put

$$I_a(\gamma, \tilde{\gamma}) = \iint \frac{\mathbf{1}_{\{\text{dist}_{\text{sph}}(x, y) < 2a\}}}{a^2} d\ell_\gamma(x) d\ell_{\tilde{\gamma}}(y).$$

If the two paths meet at $x \in U \cap G$, then for every sufficiently small a the physical ball of radius a about x is contained in $U \cap G$. Each path has at least $a/2$ of arclength in its ball of radius $a/2$: after visiting x it must eventually leave the ball in order to reach its dust endpoint. Therefore

$$\liminf_{a \downarrow 0} I_a(\gamma, \tilde{\gamma}) \geq \frac{1}{4}$$

on the domain-intersection event. The spherical kernel in this integral has uniformly bounded integrals in each variable, since spherical balls of radius $2a$ have area at most a numerical constant

times a^2 . Its operator norm on $L^2(da)$ is consequently bounded by a numerical constant C . Independence yields

$$\mathbb{E} I_a \leq C \|g_{\mu_p}\|_{2,U \cap G} \|g_{\mu_q}\|_{2,U \cap G} \leq C \delta^2.$$

Applying Fatou's lemma along any sequence $a \downarrow 0$ gives

$$\mathbb{P}(\text{a common visit in } U \cap G) \leq 4C \delta^2. \quad (6.3)$$

The estimates concern the original paths, and hence also bound intersections of their last-hit tails. *The contradiction.* Combining Equations (6.2) and (6.3), and taking the union bound over the four cross-pairs, bounds the alternation event by

$$4(2\delta + (1 + 4C)\delta^2).$$

For sufficiently small δ this is less than 10^{-4} , the contradiction. $\square$

6.3 Selecting barriers and collapsing unmarked components

The preceding theorem concerns availability for a fixed prior list. We use it after a maximal countable selection, so that every point still uncovered is unavailable for the final list. Transfinite arguments also occur in He and Schramm's treatment of countably connected domains, where the induction follows the Cantor–Bendixson complexity of the boundary-component space [7, p. 373 and Theorem 7.1]. Here each new test enlarges an open set of ends already supplied with a barrier, and second countability limits the number of strict enlargements. The whole space of complementary components need not be countable.

Start with an arbitrary countable well-ordered compatible list $\mathcal{L}_0$ and its countable flag set Σ_0 . Append ungated real barriers for the anchor in Theorem 6.3, without changing the earlier coordinates, masks, periods, or budgets. The seed list may be empty. During the selection, let V_k be the relatively open subset of B covered by the inner dust disks of the newly chosen barriers with norm bound strictly less than $1/k$, for each $k \in \mathbb{N}$. At a countable stage, if some pair (k, b) with $b \notin V_k$ admits a compatible barrier with that bound, append one such barrier and enlarge the flags as permitted. Its inner dust disk strictly enlarges V_k . At countable limit stages take unions of the lists and of the flags. Compatibility persists by its finite-sublist formulation.

This process stops at a countable stage. Indeed, fix a countable basis for B . At a step enlarging V_k , choose a basis element containing the previously uncovered endpoint and contained in the new inner dust disk. It is now contained in V_k and was not contained in V_k before this step. The pair consisting of k and this basis element can never be chosen again. An ω_1 -long sequence of steps would therefore inject into a countable set, which is impossible.

Let $\mathcal{L}$ be the resulting countable compatible list and set

$$S = \bigcup_{k \in \mathbb{N}} (B \setminus V_k). \quad (6.4)$$

At termination no endpoint in $B \setminus V_k$ admits an extension with bound less than $1/k$ for the final list and its final flags. Theorem 6.3, applied separately for each k , shows that S is countable. We henceforth include all its points among the mandatory marks in the finite transfers.

It remains to convert the selected barriers into a statement about limiting components. The following estimate pays for a change from zero to one along a line by the domain gradient and the oscillations across the round holes it meets.

Lemma 6.4 (Payment along parallel lines). *Let D be a finite circle domain containing infinity, with all its bounded complementary disks contained in $\{|w| < L\}$. Write their radii as r_T , allowing radius zero. Let v be a smooth real function on D of finite energy. Suppose its cluster values near each complementary disk lie in a bounded real interval of length at most $A(T)$.*

Fix a direction and a measurable set J of transverse parameters. Suppose that on each line with parameter in J there is a segment in $\{|w| \leq L\}$ with endpoints in D on which v has values zero and one. Then

$$|J| \leq \sqrt{\pi} L \|dv\|_{2,D \cap \{|w| \leq L\}} + 2 \sum_T r_T A(T) \leq \sqrt{\pi} L \|dv\|_2 + 2L \left(\sum_T A(T)^2 \right)^{1/2}. \quad (6.5)$$

The same first inequality holds with the gradient integrated over any measurable region containing all the chosen domain portions of the segments.

Proof. Discard the finitely many tangent lines and lines through point holes. For almost every remaining line the gradient integral in the bounded region is finite. The intersection of a chosen segment with D is a finite union of intervals. On each, ordinary variation bounds the change of v by the gradient line integral. Approach the endpoints of the complementary intervals from within D . The difference across each such interval is at most $A(T)$, by the cluster-interval hypothesis. Equivalently one can first use compact subintervals in D and then take endpoint limits; finite line variation supplies the limits whenever needed. Thus

$$1 \leq \int_{D \cap \text{segment}} |dv| |dw| + \sum_{T: T \cap \text{segment} \neq \emptyset} A(T).$$

Majorizing by the full-line gradient integral in the radius- L ball and by all disk hits makes this bound independent of any choice of segments. Integration in the transverse parameter and Cauchy–Schwarz give the first inequality in Equation (6.5). A round disk of radius r_T is met by a set of parallel lines of parameter length $2r_T$. Finally the interiors of the disks are disjoint and lie in the radius- L ball, so

$$\sum_T r_T^2 \leq L^2, \quad \sum_T r_T A(T) \leq L \left(\sum_T A(T)^2 \right)^{1/2}.$$

The proof uses the gradient only on the actual domain portions of the segments, which also proves the localized assertion. $\square$

Theorem 6.5 (Collapse outside the first exceptional set). *Let $\mathcal{L}$ and S be obtained by the preceding selection, starting with any countable compatible seed list. Extend $\mathcal{L}$ by any further countable compatible list, allowing further countable flag enlargements, and take a diagonal finite-transfer limit f as in Section 3. Then $K_b(f)$ is a singleton for every $b \in B \setminus S$.*

Proof. Fix $b \notin S$. If $K_b(f)$ had a projection of positive length, every selected barrier surrounding b would have to pay for that same length. Their arbitrarily small norms will contradict this requirement.

Suppose $K_b(f)$ is nondegenerate. Some orthogonal projection of this compact continuum has positive length. Fix a compact interval J of positive length inside that projection. The target bound L from Section 3 can be increased once to contain all images of the outer zero circle E_* as well as all relevant inner curves and complementary disks. This choice is independent of the barrier: the inner curves lie in the bounded side of the fixed outer curve, whose image is uniformly bounded.

For each k choose a selected barrier u with

$$\|du\|_2 + \|a_u\|_{\ell^2} < 1/k$$

whose inner dust disk contains b . Its inner boundary curve C has value one, and E_* has value zero. The end-component description in Lemma 3.3 places $K_b(f)$ strictly inside the bounded Jordan region of $f(C)$, which in turn lies inside that of $f(E_*)$. For this fixed test, local uniform convergence on the two curves and stability of winding imply the same inclusions of the fixed compact set $K_b(f)$ for all sufficiently large finite maps f_n .

Each line with parameter in J therefore meets the inner bounded region. Following it out toward infinity supplies a segment between $f_n(C)$ and $f_n(E_*)$, with test values one and zero. Both endpoints, and hence the segment, lie in the radius- L ball. Apply Lemma 6.4 to the transferred test on the finite target circle domain. Its energy norm has upper limit at most $\|du\|_2$, since it agrees with u on exhausting cores and its remaining energy tends to zero. Its boundary oscillations satisfy

$$A_n(T) \leq a_u(s_T) + e_n(T), \quad e_n(T) \geq 0, \quad \sum_T e_n(T)^2 \rightarrow 0.$$

The terminal indices s_T are distinct. Consequently

$$\limsup_n \left(\sum_T A_n(T)^2 \right)^{1/2} \leq \|a_u\|_{\ell^2(B)}.$$

Taking the limit in Equation (6.5) gives

$$|J| \leq \sqrt{\pi} L \|du\|_2 + 2L \|a_u\|_{\ell^2} \leq \frac{(\sqrt{\pi} + 2)L}{k}.$$

The interval J is fixed and k is arbitrary, a contradiction. Further list extensions preserve both the old tests and their compatibility, so the argument applies to every stated extension. $\square$

6.4 Free contacts with prior levels

The first selection has now reduced the component problem to the countable set S . To construct period coordinates at those ends, we need a further property of the path laws. The residual level constraint already controls residual dust contacts; the next lemma extends that control to contacts inside tags and obtains a finite expected count of all intersections with a generic prior level union.

Lemma 6.6 (Generic level intersections of a path law). *Fix a countable prior compatible list with flag set Σ , possibly containing period coordinates, and let μ be a law from Theorem 5.1. For every prior coordinate i , almost surely the trace values modulo one at the free points of the path in $B \cap M_i$ form a Lebesgue-null subset of $\mathbb{T}$.*

Consequently, for almost every tuple defining any fixed finite prior level union H ,

$$\mathbb{P}_\mu\{\gamma([0, 1]) \cap H \cap (B \setminus \Sigma) = \emptyset\} = 1, \quad \mathbb{E}_\mu \#(\gamma([0, 1]) \cap H) < \infty.$$

For one coordinate the integrated distinct-intersection bound is

$$\int_{\mathbb{T}} \mathbb{E}_\mu \#(\gamma([0, 1]) \cap H_i(t)) dt \leq \int_{G \cap M_i} g_\mu |du_i| da + \sum_{s \in B} m_\mu(s) a_i(s) < \infty. \quad (6.6)$$

Proof. We first show that, at a generic independent level, free contacts inside a tag can occur only in singleton level components. A detour around such singleton contacts then gives a variation estimate for the prior coordinate along the tag. That estimate makes the free trace image null. Finally, coarea counts the remaining intersections.

Singleton contacts at an independent level. Fix a tag slot, its type, and a prior coordinate i . For almost every pair consisting of the tag tuple and an independent level t , every free contact of that tag with $H_i(t)$ has a singleton component in $H_i(t)$. To prove this, exclude equality between t and every level of label i in the tag, and impose the finitely many relevant instances of (P). If a level of some tag label $j > i$ is present at a contact x , (P) for j against i gives the assertion. If no such higher label is present, the tag yields a nondegenerate local continuum through x in labels lower than i , exactly as in the proof of Lemma 6.2. This is forbidden by (P) for i . Closedness permits levels absent at x to be removed locally. Absolute continuity of the tuple law in each slot and type, followed by Fubini, shows that for almost every sample this singleton assertion holds for almost every t , simultaneously for all its tags.

A local variation bound. Near a free point in $B \cap M_i$, choose an open topological disk V contained in the interior of the mask, away from the singularity if there is one. For a period coordinate choose a common lift on V . Shrink V until all the defined values and bands of this branch lie in one real interval of length less than $1/2$. Such a choice is possible because the point has a unique trace. A countable family of these disks covers the relevant free points.

Consider two free contact times $r < s$ in one tag such that $\gamma([r, s]) \subset V$. Let their real trace values be α and β . We claim that

$$|\beta - \alpha| \leq \int_{[r, s] \cap \gamma^{-1}(G \cap M_i)} |du_i|(\gamma) ds + \sum_{t \in [r, s]: \gamma(t) \in \Sigma \cap M_i} a_i(\gamma(t)). \quad (6.7)$$

Here the sum again counts parameter times. Assume $\alpha < \beta$. For almost every real threshold $v \in (\alpha, \beta)$ the preceding singleton assertion holds at its modulo-one level. We show that the segment must hit this threshold either at a domain point or in a flagged band.

Otherwise the compact set

$$F_v = \gamma([r, s]) \cap H_i(v \bmod 1)$$

consists only of free points that are singleton components of the closed level. They remain singleton components after adjoining B , by the dust topology of Section 2. The small-disk construction of Lemma 2.2 covers F_v by finitely many open disks with closures in V , missing the endpoints and with boundaries avoiding the entire closed level and B . Process these disks successively on the current path segment, skipping any whose interior is no longer met. Replace the portion from first entry to last exit of a disk by an arc of its boundary. Each replacement removes the current level hits in that disk and adds none, so after the finite process all level hits have disappeared. Disks need not be disjoint for this argument.

The resulting path in V cannot exist. Off the level, the sign of the chosen branch relative to v is defined at domain points and at dust points by their bands, and is locally constant. At dust points this follows from the defining upper and lower limits of the band. The endpoint signs are opposite. The local range of length less than $1/2$ ensures that no other integer translate of the level is involved. This proves the required domain-or-flag hit for almost every intervening threshold.

The set of values contributed by the domain times has length at most the integral in Equation (6.7): exhaust the countably many domain intervals and apply ordinary curve variation on their rectifiable physical portions. The flagged bands contribute at most the sum in that equation.

These ranges cover almost every threshold between α and β , proving the claim. The case $\beta < \alpha$ follows by reversing the endpoints.

Null trace image. For almost every sample, the right-hand costs above are the restrictions of a finite measure κ_i on path time: its domain part is $|du_i| ds$ within the mask, and it has an atom of mass $a_i(\gamma(t))$ at each flagged visit time. Indeed

$$\mathbb{E}_\mu \kappa_i([0, 1]) \leq \|g_\mu\|_2 \|du_i\|_{2, G \cap M_i} + \|n_\mu|_\Sigma\|_{\ell^2} \|a_i\|_{\ell^2} < \infty.$$

The measure gives zero mass to free contact times. Fix a tag and one of the disks V . By outer regularity, its free contact times in $\gamma^{-1}(V)$ have an open cover, within that tag and $\gamma^{-1}(V)$, of arbitrarily small κ_i -measure. The cover has countably many interval components. Within each component, Equation (6.7) bounds the diameter of the free trace image by the measure of that component. Summing proves that the free trace image in this tag and this disk has length zero. The disks and tags are countable.

On residual times, the same null-image conclusion follows directly from the residual level constraint by deleting the flags, as in the proof of Lemma 6.2. Thus all free contacts have null trace image. Singular points are flagged and play no role in this assertion.

Distinct intersection counts. The image of the domain times is countably rectifiable, and its physical length measure is dominated by the arclength occupation. The one-dimensional area formula applied to the smooth map u_i modulo one, in local physical charts and lifts, bounds the integrated number of distinct domain intersections by

$$\int_{G \cap M_i} g_\mu |du_i| da.$$

This also applies at mask boundaries by local smoothness; the occupation of each such fixed smooth boundary is zero almost surely because its area is zero. At a dust point, the parameter length of level incidence is at most $a_i(s)$. The already proved null-image assertion removes free incidences under the joint law and level measure. On the countable flags, Tonelli gives the remaining bound $\sum_s m_\mu(s) a_i(s)$, including a possible singular point with its budget one. Cauchy–Schwarz makes both bounds finite and proves Equation (6.6).

For a finite union, sum these bounds over its finitely many labels and apply Fubini to its level tuple. The null free-incidence assertion also passes to that union. Incidence and distinct finite-point counts are measurable in the completed sense supplied by the path coding, so these integrations are legitimate. $\square$

7 Period coordinates and the marked components

The first selection has made every unmarked complementary component a point. It remains to show that, at each marked end, the limiting round disk fills the entire complementary component. Suppose that part of the component protrudes beyond that disk. A sufficiently small real barrier excludes the protrusion by the line-payment estimate. If no such barrier exists, an endpoint law produces a coordinate with additive period one. Before smoothing, its boundary variation is a probability measure; its energy and its oscillations at all other ends can be made arbitrarily small. These properties give incompatible lower and upper bounds for a boundary integral, the upper bound coming from Stokes' theorem.

Throughout this section the prior coordinate list is an arbitrary countable list satisfying (P). All references to free points use its current flag set, enlarged when a new coordinate is appended.

7.1 Reduction to a compact family of simple arcs

We first retain only the terminal portions of the endpoint paths and erase their loops. Restricting the resulting law to a compact family will give the uniform winding bounds needed for the period coordinate.

Lemma 7.1 (Compact simple-arc reduction). *Let μ be an endpoint law supplied by Theorem 5.1, with target b and prescribed collars $C_j(b) = \partial U_j(b)$. For every sufficiently deep prescribed collar $C = \partial U$, there is a probability law η supported on a compact subset of $C([0, 1], X)$ consisting of simple arcs α such that*

$$\alpha(0) \in C, \quad \alpha(1) = b, \quad \alpha((0, 1)) \subset U \setminus \{b\}.$$

Its mean domain occupation has a density g_η satisfying

$$g_\eta \leq 2\mathbf{1}_{U \cap G} g_\mu \quad \text{almost everywhere,} \quad m_\eta(s) \leq 2m_\mu(s) \quad (s \in B \cap \bar{U}), \quad (7.1)$$

and $m_\eta = 0$ outside $\bar{U}$. Here m denotes hit probabilities, rather than visit counts. In particular, for every $\varepsilon > 0$, every sufficiently deep prescribed collar admits such a law η with

$$\|g_\eta\|_2 + \|m_\eta|_{B \setminus \{b\}}\|_{\ell^2} < \varepsilon. \quad (7.2)$$

For almost every tuple defining any fixed finite prior level union H , the expected number of distinct intersections of an arc with H is finite, and the arcs almost surely have no free dust contacts in H . If the endpoint problem has gates, the starting point additionally satisfies $f_0(\alpha(0)) \in \bar{W}_2$.

Proof. Cut each path at its last hit of C and retain its terminal part, reparametrized on $[0, 1]$. The anchor is outside $\bar{U}$, so such a hit exists. Its last hitting time is strictly less than 1; after that time the path is strictly inside U . The original path reaches b for the first time at its endpoint. Thus its terminal part has distinct endpoints and contains a simple arc with those endpoints. The gate condition passes to the last hitting point.

These operations can be performed measurably up to completion of the law. Last hitting times of a fixed closed set are Borel on the present path family. In the product of two path spaces, the conditions that the second path is injective, has prescribed endpoints, and has image contained in the first path's image are Borel. For injectivity, use

$$\min_{|s-t| \geq 1/n} d_X(\alpha(s), \alpha(t)) > 0 \quad (n \in \mathbb{N});$$

image containment is closed for uniform convergence. The nonempty sections therefore admit a universally measurable selection by the Jankov–von Neumann theorem [10, Theorems 18.1 and 29.9]. The pushforward of the completed probability measure is an ordinary Borel probability measure on path space.

The occupation measure of a selected simple arc is its domain image length measure. By the image-length observation in Section 4, this is dominated by the occupation of the original terminal path. There is no arclength atom at its initial point on C . Consequently the unconditioned selected-arc law has occupation density at most $\mathbf{1}_{U \cap G} g_\mu$, and its hit probabilities are at most the original hit probabilities. The endpoint and interior conditions describe a Borel subset of path space. Inner regularity provides a compact subset of that set of probability at least $1/2$. Conditioning on this subset proves Equation (7.1).

The indicators of $U_j(b) \cap G$ decrease pointwise to zero. The square-summable function m_μ has countable support, and the sets $(U_j(b) \cap B) \setminus \{b\}$ decrease to the empty set. Absolute continuity of the two squared integrals therefore proves Equation (7.2). Finally, all selected images are subsets of the original images. The level-intersection conclusions of Lemma 6.6 survive restriction, selection, and conditioning, with their expected counts increased by at most the conditioning factor. $\square$

7.2 Averaging the index of a ray

Each simple arc will determine an integer-valued function on the cyclic cover of the punctured dust disk. We average these functions to obtain a continuous period coordinate, then smooth it while preserving all nonsingular bands.

Lemma 7.2 (A stream coordinate). *Let U be a dust disk, let $b \in B \cap U$, and put $C = \partial U$. Suppose η is supported on a compact family of simple arcs from C to b , meeting C only at their initial points and b only at their terminal points. Assume that its mean domain occupation has an L^2 density g_η , that $m_\eta \in \ell^2(B)$, and that it has the generic level-intersection conclusions of Lemma 7.1 for the prior list.*

There is a continuous, locally Sobolev function v on the cyclic cover over $G \cap \overline{U}$, with additive period one about b , such that

$$\|dv\|_{2,G \cap U} \leq \sqrt{2} \|g_\eta\|_2. \quad (7.3)$$

At every $s \in B \cap U \setminus \{b\}$ its band is bounded and has width at most $m_\eta(s)$. After enlarging the flags by $\{s \in B : m_\eta(s) > 0\}$, its free traces satisfy (P) against the prior list. If β is the probability distribution of the arc starts on the positively oriented curve C , then β is atomless and

$$d(v|_C) = \beta. \quad (7.4)$$

This equality is understood on a lifted boundary interval: the branch increases by one over a full turn.

For every $\varepsilon > 0$ one may replace v by a smooth coordinate u of the same period such that

$$\|d(u - v)\|_{2,G \cap U} < \varepsilon, \quad \sup_C |u - v| < \varepsilon, \quad (7.5)$$

with the same nonsingular bands and hence with (P). The discrepancy $u - v$ is single-valued. Thus u is an admissible coordinate with mask $\overline{U}$ and budget

$$a_u(s) = m_\eta(s) \quad (s \in B \cap U \setminus \{b\}), \quad a_u(b) = 1,$$

and zero budget off its mask.

Proof. We first construct v and prove its energy estimate and boundary-measure identity. We then control its dust bands and free traces, and finally produce the smooth admissible coordinate u .

Step 1: the lifted rays. Choose an orientation-preserving homeomorphism from $(\overline{U}, b)$ to the closed unit disk with its center. Write the cyclic cover of $\overline{U} \setminus \{b\}$ as

$$\mathcal{H} = \mathbb{R} \times [0, \infty),$$

where the first coordinate is angle divided by 2π , the second is minus logarithmic radius, and the deck transformation is $(x, y) \mapsto (x + 1, y)$. These are topological coordinates; all analytic estimates below will instead use physical conformal charts in G .

For an arc α , choose the lift starting at $(a, 0)$ with $a \in [0, 1)$. Its image Γ_α is a proper simple ray with $y \rightarrow \infty$ at its terminal end. Its integer translates are disjoint: an intersection of two translates would give a repeated projected point of the simple arc. In the one-point compactification of $\mathcal{H}$, the translated ray $\Gamma_\alpha + (k, 0)$ is a crosscut from $(a + k, 0)$ to the boundary point at infinity. Let $R_k(\alpha)$ be its right-hand complementary component, the one incident to boundary points $(x, 0)$ with $x > a + k$. Jordan separation gives

$$R_{k+1}(\alpha) \subset R_k(\alpha).$$

Indeed the next ray lies on the right of the preceding ray because it is connected, is disjoint from that ray, and starts on its right. The right side of the next ray connects to boundary points right of both starts and is therefore contained in the preceding right side.

Step 2: the integer index and its local bounds. Compactness of the arc family has two consequences. The arcs approach b uniformly as the parameter tends to 1; and, for every $\delta > 0$, their portions with parameter in $[0, 1 - \delta]$ stay a uniform positive distance from b . The first assertion follows from equicontinuity and the common terminal endpoint. The second follows from compactness and the fact that every member is injective and meets b only at parameter 1.

Fix $0 \leq Y < \infty$. Uniform approach to b gives $\delta > 0$ such that every arc point with $0 \leq y \leq Y$ occurs before the common time $1 - \delta$. On this initial interval the arcs stay uniformly away from b , so their projected circular positions are equicontinuous. Subdivide time into finitely many intervals on each of which all circular positions move by less than a semicircle. Starting with $a \in [0, 1]$, successive lifts then give a uniform bound for all lifted angles on this initial interval. Thus all unshifted rays in the strip $0 \leq y \leq Y$ lie in a common bounded horizontal interval.

For a compact subset of $\mathcal{H}$, join its points vertically to the boundary. Sufficiently large positive or negative translates miss all those vertical segments. Their side at the boundary is fixed by the order of the starts, and it remains fixed along the segments. Hence, for every z off the translated rays, the set $\{k \in \mathbb{Z} : z \in R_k(\alpha)\}$ is nonempty and bounded above. We may therefore define

$$J_\alpha(z) = \max\{k \in \mathbb{Z} : z \in R_k(\alpha)\}. \quad (7.6)$$

The same argument bounds $|J_\alpha|$ uniformly over the arc family on each compact subset of $\mathcal{H}$, wherever it is defined. The definition also gives

$$J_\alpha(z + (1, 0)) = J_\alpha(z) + 1, \quad J_\alpha(x, 0) = \lfloor x - a \rfloor \quad (x - a \notin \mathbb{Z}). \quad (7.7)$$

Step 3: measurability and an intersection inequality. The lifted arc on every initial compact time interval is a Borel function of the arc and its chosen starting angle, by path lifting. Off a ray, membership in its right component is Borel as well. One may test it by countable chains of rational open balls disjoint from the ray and joining the point to the appropriate boundary interval. Avoidance of a compact test path is Borel by the initial lifts and a countable exhaustion of $[0, 1]$. Since a compact path disjoint from a closed proper ray has positive distance from it, these tests detect all complement connections. This proves measurability of the indices. On the coincidence set, where z lies on a translated ray, assign an arbitrary value whenever a random variable is required. We will take expectations only at points whose coincidence probability is zero.

Suppose a curve δ in the cover projects injectively to $\bar{U} \setminus \{b\}$ and its endpoints miss the sample rays. Every ray separating the endpoint indices must meet δ . Different such meetings project to different points because the projection of δ is injective. Consequently

$$|J_\alpha(\delta(1)) - J_\alpha(\delta(0))| \leq \#(\alpha([0, 1]) \cap \text{proj } \delta([0, 1])). \quad (7.8)$$

The right side counts distinct projected points; the inequality also holds when that count is infinite.

Step 4: averaging and continuity. The index is now defined, measurable, and locally uniformly bounded. We use these properties to average it. Lemma 6.1 applies to η , since its arcs end at $b \in B$ and have an L^2 mean occupation density. Thus every fixed point of $G \cap \bar{U}$, including a point of C , is missed almost surely. For each lift z of such a point, define

$$v(z) = \mathbb{E}_\eta J_\alpha(z).$$

Local uniform domination makes this expectation finite. For a convergent sequence in the lifted domain, discard the null coincidence events at its terms and its limit. For every remaining sample the index is locally constant at the limit. Dominated convergence proves continuity of v , including at C . The first identity in Equation (7.7) gives additive period one.

Step 5: the differential estimate. Work in a physical conformal rectangle compactly contained in $G \cap U$, with a common lift. Here $z = x + iy$ denotes a physical conformal coordinate, and the spherical metric is $\lambda(z)|dz|$; these x, y are unrelated to the topological angle and depth above. For a horizontal interval I and height y let $N_\alpha(I, y)$ count the distinct intersections of the sample arc with $I \times \{y\}$. Curve coarea, followed by the occupation bound, gives, for Borel sets A of heights,

$$\int_A \mathbb{E}_\eta N_\alpha(I, y) dy \leq \int_{I \times A} \lambda(z) g_\eta(z) dx dy. \quad (7.9)$$

To check the conformal factor, Euclidean image length is spherical image length divided by λ , whereas spherical area is $\lambda^2 dx dy$. Projection coarea costs at most Euclidean image length. The same estimate holds after restricting counts to Borel portions of an interval.

Apply Equation (7.9) first to closed intervals with rational endpoints. Outside one null set of heights, the mean counting measure on every compact interior interval is bounded by $\lambda g_\eta dx$. This follows by the countable interval tests and then measure approximation. Averaging Equation (7.8) shows on those lines that

$$|v(x_2, y) - v(x_1, y)| \leq \int_{x_1}^{x_2} \lambda(x, y) g_\eta(x, y) dx.$$

Continuity permits arbitrary endpoints. Thus v is absolutely continuous on almost every horizontal line and $|\partial_x v| \leq \lambda g_\eta$ there. The vertical argument gives $|\partial_y v| \leq \lambda g_\eta$. The usual characterization by absolute continuity on lines yields local Sobolev regularity and

$$|dv|_{\text{sph}} \leq \sqrt{2} g_\eta \quad \text{almost everywhere.}$$

An exhaustion by charts proves Equation (7.3).

Step 6: the boundary measure. The distribution β of the starting point has no atoms by point avoidance on C . Between two ordered lifted boundary angles, the expected difference of the floors in Equation (7.7) counts the mean number of start translates in that interval. This proves Equation (7.4). In particular, for every continuous real function φ on C ,

$$\int_C \varphi dv = \int \varphi(\alpha(0)) d\eta(\alpha). \quad (7.10)$$

The left side is a Stieltjes integral. Only the orientation of the topological angular coordinate is used; no smoothness is required.

We have obtained a continuous period coordinate with the required energy bound and a positive boundary measure of total mass one. To make it admissible for transfer, we must still control its dust bands, verify compatibility, and smooth it.

Step 7: bands at the dust. Fix $s \in B \cap U \setminus \{b\}$ and a lift of a neighborhood of s . Consider two sequences of domain points tending to that lift. At all terms of both sequences the sample index is defined almost surely. For each sample, Step 2 shows that only finitely many translated rays meet a compact neighborhood of the chosen lift. Shrink to a connected neighborhood missing those rays that do not pass through the lift; their side memberships are then fixed. If the projected sample misses s , the eventual index difference is therefore zero. Otherwise simplicity leaves just one ray

through the lift. Only membership in its right-hand component can vary, so the index has at most two adjacent values. Hence the upper limit of the absolute index difference is at most one. The local uniform bound and the reverse Fatou inequality now give

$$\limsup |v(z_n) - v(w_n)| \leq m_\eta(s).$$

Local boundedness makes the upper and lower bands finite, and choosing sequences approaching their extremes proves the asserted width bound. When $m_\eta(s) = 0$, the unique trace is $\mathbb{E}_\eta J_\alpha(s)$.

Step 8: compatibility. Put $\Sigma' = \Sigma \cup \{s \in B : m_\eta(s) > 0\}$, a countable enlargement of the flags. Fix a generic finite prior level union H . By hypothesis the rule

$$\nu(A) = \mathbb{E}_\eta \#(A \cap H \cap \alpha([0, 1])), \quad A \subset H \text{ Borel},$$

defines a finite measure. It gives zero mass to $H \cap (B \setminus \Sigma')$, since almost every sample avoids this whole set. The counting integrals are measurable by compact incidence tests and finite distinct-point tests in path space; finite expected total count then defines the measure on all Borel subsets.

Let K be a nondegenerate component of H and work in a small disk V with closure in $U \setminus \{b\}$ and with a common lift. For free points $x, y \in K \cap V$ joined by a simple arc $\delta \subset K \cap V$, their traces are expectations of the indices. Equation (7.8) gives

$$|v(x) - v(y)| \leq \nu(\delta). \quad (7.11)$$

Set $F = K \cap V \cap (B \setminus \Sigma')$. We have $\nu(F) = 0$, and K is locally connected by Lemma 2.5. Equation (7.11) verifies the arc hypothesis of Lemma 2.6 for the continuum K , the relatively open set $K \cap V$, the measure $\nu|_K$, and the free set F . Hence its real trace image has Lebesgue outer measure zero. Countably many components K and lifted disks V suffice. The singular point b is flagged. Fubini in the level parameters therefore proves (P) for the new coordinate. All incidence statements may use completed measures, as in the earlier level arguments.

Step 9: smoothing. It remains to smooth v with an error that vanishes at every dust point, so that all nonsingular bands remain unchanged. Near the smooth curve C , the continuous Sobolev branches extend across the boundary by reflection in smooth collar charts. Cover $G \cap \overline{U}$ by a locally finite family of relatively compact simply lifted patches and take a smooth partition of unity. Smooth a branch on each patch; on other sheets use its integer translates. The local smoothing discrepancy is single-valued, so partition interpolation respects the additive period.

Choose a continuous positive tolerance e on the source patches with

$$0 < e(x) \leq \frac{1}{2} \min\{\varepsilon, \text{dist}_X(x, B)\}, \quad x \in G \cap \overline{U}.$$

It extends positively across the compact collar of C . On the j th patch choose the branch approximation error in values smaller than the positive infimum of e on the corresponding compact partition support. Simultaneously make its Sobolev error, including the derivatives of the partition function, less than $\varepsilon 2^{-j}$. Local continuity and Sobolev approximation permit both requirements. Partition interpolation then gives $|u - v|(x) \leq e(x)$, while the differential errors are summable. The resulting u satisfies Equation (7.5) and is smooth up to C . For every domain sequence tending in X to a nonsingular dust point, $e(x)$ tends to zero; hence the upper and lower band limits are exactly unchanged in a common local lift. Its free traces are therefore unchanged. Since compatibility for a newly appended coordinate concerns only those traces against earlier level unions, (P) is preserved. $\square$

7.3 A boundary estimate for a transferred stream

The boundary-measure identity will provide a lower bound near a protrusion. The following Stokes estimate supplies the upper bound in a finite completion, using the energy and the oscillations at the other holes.

Lemma 7.3 (Stokes estimate). *Let u be a smooth period coordinate with mask $\bar{U}$ and singularity b , and put $C = \partial U$. Transfer it to a finite completion retaining C and the marked hole indexed by b . Write F for the normalized circle-domain map, u^* for the transferred coordinate in the target, and D_b for the distinguished round hole. Assume that the masked region lies in $|w| < L$. Target lengths and areas are Euclidean; the norm below is over the domain inside $F(C)$ and outside its circular holes. For a target hole T inside $F(C)$, let s_T be the index of its source terminal disk and r_T its radius. When $s_T \neq b$, let $A_u(T)$ be its transferred boundary oscillation. Then*

$$\left| \int_{F(C)} \text{dist}(w, D_b) du^* \right| \leq \sqrt{\pi} L \|du^*\|_2 + 2\pi \sum_{T: s_T \neq b} r_T A_u(T). \quad (7.12)$$

Consequently, along diagonal transfers F_n with errors tending to zero, write u_n^* for the transferred coordinate and $D_{n,b}$ for its distinguished target disk. Then

$$\limsup_{n \rightarrow \infty} \left| \int_{F_n(C)} \text{dist}(w, D_{n,b}) du_n^* \right| \leq \sqrt{\pi} L \|du\|_{2, G \cap U} + 2\pi L \|a_u|_{B \setminus \{b\}}\|_{\ell^2}. \quad (7.13)$$

Proof. Put $\phi(w) = \text{dist}(w, D_b)$. Its weak gradient has norm at most one. Apply Stokes' theorem in the region inside $F(C)$ after replacing each circular hole by a slightly larger concentric circle. There are only finitely many holes, so the offsets can be chosen disjoint and inside the domain. The exterior boundary integral is the wedge integral $\int d\phi \wedge du^*$ plus the integrals around the offset circles, each with positive disk orientation.

On the distinguished offset of radius $r_b + \rho$, ϕ is the constant ρ . The period of u^* there is one, so its contribution is ρ and tends to zero.

At every other hole the period is zero, so there is a single-valued branch. Subtracting a constant from that branch and integrating by parts bounds the boundary integral by

$$2\pi(r_T + \rho)(A_u(T) + o(1)).$$

Here $|d\phi| \leq |dw|$ along the Euclidean offset circle, and the oscillation on small offsets tends to at most $A_u(T)$ by the collar correspondence in Lemma 3.3. This argument also covers a hole of radius zero. The remaining region has Euclidean area at most πL^2 , so Cauchy–Schwarz bounds the wedge integral by $\sqrt{\pi} L \|du^*\|_2$. Letting the offsets tend to zero proves Equation (7.12).

Disjointness of the disk interiors gives $\sum_T r_T^2 \leq L^2$. The terminal indices are distinct, and the index b occurs only at the distinguished hole. The positive-excess estimate of Theorem 3.1 therefore gives

$$\left(\sum_{T: s_T \neq b} A_u(T)^2 \right)^{1/2} \leq \|a_u|_{B \setminus \{b\}}\|_{\ell^2} + o(1).$$

Conformal invariance of Dirichlet energy and the energy estimate of that theorem also give $\|du^*\|_2 \leq \|du\|_{2, G \cap U} + o(1)$. A second application of Cauchy–Schwarz proves Equation (7.13). $\square$

7.4 Excluding a protrusion

We now apply the barrier alternative and the Stokes estimate to a fixed normalized map and a round disk properly contained in one of its marked components. The estimates will exclude a whole neighborhood of that pair, allowing the final selection to use only countably many tests.

For a fixed mark b , let $\mathcal{Q}_b$ be the space of pairs (f, D) where f is a normalized univalent map on G and D is a closed round disk, possibly a singleton, contained in $K_b(f)$. Give maps the compact-open topology and disks their center-radius topology, equivalently the Hausdorff topology on the bounded disks considered here. These pair spaces are second countable. A pair is *bad* if $D \neq K_b(f)$.

Lemma 7.4 (Local exclusion of a bad pair). *Given a countable prior list satisfying (P), a mark b , and a bad pair $(f_0, D_0) \in \mathcal{Q}_b$, one can append a compatible real barrier or a compatible period coordinate singular at b such that an entire relative neighborhood of (f_0, D_0) is excluded as limiting pair data of diagonal finite transfers. The exclusion persists after any further countable extensions satisfying (P), provided b remains marked.*

Proof. We first fix geometry that will serve both alternatives. In the barrier case it gives a family of line segments with a fixed positive transverse width. In the law case it keeps the arc starts a fixed positive distance from D_0 .

Fixed geometry near the protrusion. Write $K = K_b(f_0)$. There is a boundary point x of K outside D_0 : for example, maximize distance to the center of D_0 on K . Choose a bounded open ball W_2 about x whose closure misses D_0 , and a smaller closed ball W_1 about x contained in W_2 . Boundary bumping, applied in a still smaller closed ball, gives a nondegenerate subcontinuum $Q \subset K \cap \text{int } W_1$. The ball can be chosen small enough that K also has points outside it, because $D_0 \subset K$ is separated from x .

Every boundary point of a complementary component is approached from the domain. Indeed, if a ball about such a point missed the domain, the connected ball would lie in that same complementary component, contradicting that the point was a boundary point. Thus choose a small closed ball $A \subset f_0(G) \cap \text{int } W_1$ and nested smaller balls strictly inside A . Their preimages give an anchor E and a compact testing set $E_* \subset \text{int } E$. Start the collar sequence deep enough that its closed disks miss E .

There is a direction and a closed interval J of positive transverse width such that every line with transverse parameter in J joins a point strictly inside $f_0(E_*)$ to a point of Q , by a segment in $\text{int } W_1$. To see this, first direct a line from the center of the smallest anchor ball to a point of Q . That point projects strictly inside the ball's transverse projection. Perturb the direction, if necessary, so that the transverse projection of Q is nondegenerate. The projection is an interval, and it overlaps the interior of the anchor-ball projection in positive length. Take J compactly inside that overlap. Choose the starts on the middle chords of the anchor ball; they then have a uniform interior buffer. Convexity of W_1 puts the segments inside it.

These sets and J are fixed before the test's collar and penalty region are chosen. For every prescribed collar C_l , the compact continuum Q lies strictly inside the bounded side of $f_0(C_l)$, whereas the buffered starts lie outside. After a collar is fixed these inclusions persist under sufficiently small changes of the map, by winding number on C_l and on the boundary of E_* . In particular the buffered starting points remain images of points in E_* .

The barrier case. Use the endpoint problem with these anchors and with gates prescribed by (f_0, W_1, W_2) . Let c_L be a constant large enough for the line-payment estimate in the common

target size bound. Fix $\tau > 0$ with $c_L \tau < |J|/2$. If there is a compatible barrier u with

$$\|du\|_{2,G \setminus O} + \|a_u\|_{\ell^2} < \tau,$$

append it. It equals zero on E_* and one on some fixed inner collar C_l . In sufficiently close finite-map data the segments just constructed start in the zero region and reach the side inside $F(C_l)$; stop each segment at its first hit of that curve. The compact set $\bar{O}$ has image disjoint from W_1 under f_0 , and this separation persists under convergence on $\bar{O}$. Thus all gradients paid along these segments lie off the image of O .

Lemma 6.4, with the transfer error tending to zero, gives

$$|J| \leq c_L (\|du\|_{2,G \setminus O} + \|a_u\|_{\ell^2}) + o(1),$$

a contradiction to $c_L \tau < |J|/2$. To justify the restricted energy estimate, retain $\bar{O}$ in the unchanged core and use the original energy off O there, together with the transfer error on the rest. All required separations involve fixed compact sets after u , C_l , and O have been chosen. This excludes a relative neighborhood of f_0 , and hence a relative neighborhood of (f_0, D_0) .

The law case. If no such barrier is available, Theorem 5.1 provides an endpoint law μ obeying all prescribed gates. We will make the Stokes upper bound small while keeping the boundary integral bounded below. Put

$$d_0 = \text{dist}(\bar{W}_2, D_0) > 0.$$

Apply Lemma 7.1 at a sufficiently deep collar $C = \partial U$, and then Lemma 7.2. The resulting smooth period coordinate u can be chosen so that

$$\sqrt{\pi} L \|du\|_{2,G \cap U} + 2\pi L \|a_u|_{B \setminus \{b\}}\|_{\ell^2} < d_0/4. \quad (7.14)$$

Indeed the energy norm and the nonsingular budget norm tend to zero with the depth. The singular budget is one and is absent from the Stokes upper bound. The gate puts all arc starts in $f_0^{-1}(\bar{W}_2)$, so the unsmoothed coordinate satisfies

$$\int_C \text{dist}(f_0(\cdot), D_0) dv = \int \text{dist}(f_0(\alpha(0)), D_0) d\eta(\alpha) \geq d_0. \quad (7.15)$$

We next preserve this lower bound while smoothing. On the smooth compact curve C , the function $\psi = \text{dist}(f_0(\cdot), D_0)$ is Lipschitz. Since $u - v$ is single-valued on C , integration by parts for a continuous function of bounded variation gives

$$\left| \int_C \psi d(u - v) \right| = \left| \int_C (u - v) d\psi \right| \leq \sup_C |u - v| \text{Var}_C(\psi).$$

After the collar is fixed, choose the smoothing accuracy so fine that this error is less than $d_0/4$, while maintaining Equation (7.14). Thus

$$\int_C \text{dist}(f_0(\cdot), D_0) du > 3d_0/4.$$

For fixed smooth $u|_C$, this integral varies continuously with the map uniformly on C and with the disk in Hausdorff distance. Consequently it remains greater than $d_0/2$ on a relative neighborhood $\mathcal{V}$ of (f_0, D_0) .

Append u and retain b as a mark. In a finite transfer, the transferred coordinate agrees with u on C for all sufficiently late stages. If the limiting pair belonged to $\mathcal{V}$, the outer integral in

Lemma 7.3 would therefore have $\liminf$ greater than $d_0/2$. Equations (7.13) and (7.14) give a $\limsup$ less than $d_0/4$. This contradiction excludes $\mathcal{V}$.

Persistence. In either case the contradiction uses one fixed appended coordinate, fixed compact source sets, its fixed energy and band bounds, and the vanishing errors for its eventual transfers. Enlarging the coordinate list or the flag set does not change these data. Later transfers still satisfy all the estimates once this coordinate and the required mark are included. The exclusion therefore persists under all further compatible countable extensions. $\square$

7.5 Selection with prescribed retained tests

The two selection arguments can start with coordinates already supplied by an application. We record this stronger conclusion explicitly: the finite-transfer estimates for those coordinates survive every later selection, and the limit still has a circle-domain image.

Theorem 7.5 (Retained-test transfer and selection). *Use the quotient sphere $X = G \cup B$, fixed exterior radius R , and coordinates of Definition 2.3. Let $\mathcal{L}_0$ be any countable well-ordered list satisfying compatibility (P), with a countable flag set Σ_0 . Prescribe also a countable set $S_0 \subset B$ of mandatory marks containing every singularity of $\mathcal{L}_0$.*

There are a countable well-ordered compatible extension $\mathcal{L}$ of $\mathcal{L}_0$, with enlarged countable flags, and a countable mark set $S \supset S_0$ containing all its singularities, with the following properties. Every normalized diagonal finite-transfer limit of $\mathcal{L}$, retaining the marks S , has a circle-domain image. This conclusion persists after any further countable compatible extension, retaining S and including all additional singularities among the marks.

More precisely, let $\varepsilon_n > 0$ tend to zero and let $H_n \subset G$ be increasing compact sets such that every compact subset of G is contained in $\text{int } H_n$ for all sufficiently large n . One can choose increasing finite subsets I_n exhausting the labels of $\mathcal{L}$, finite bordered completions (Q_n, g_n) , nested smooth cores $K_n \subset G$, transferred coordinates $u_{i,n}$, and normalized conformal maps

$$f_n : Q_n^\circ \longrightarrow \Omega_n, \quad f_n(z) = z + O(1/z),$$

where Ω_n is a finite circle domain, such that $H_n \subset \text{int } K_n$ and the following assertions hold.

- (i) *The metric g_n is the original spherical metric near K_n . For every $i \in I_n$, the coordinate $u_{i,n}$ agrees with u_i near $K_n \cap M_i$, and all its terminal periods are those specified in Theorem 3.1.*
- (ii) *The total additional energy satisfies*

$$\sum_{i \in I_n} \|du_{i,n}\|_{2, (Q_n \setminus K_n) \cap M_i, g_n}^2 < \varepsilon_n^2. \quad (7.16)$$

For every $i \in I_n$, if $T_{n,s} \subset M_i$ is a terminal disk and s is nonsingular for u_i , its boundary oscillation $A_{i,n}(T_{n,s})$ satisfies

$$\left(\sum_{\substack{s: T_{n,s} \subset M_i \\ s \neq p_i}} [A_{i,n}(T_{n,s}) - a_i(s)]_+^2 \right)^{1/2} < \varepsilon_n. \quad (7.17)$$

The singularity exclusion is omitted for real coordinates.

- (iii) *Every prescribed mark eventually indexes a terminal disk. All terminal disks have quotient diameter tending uniformly to zero. After passing to a subsequence, $f_n \rightarrow f$ locally uniformly on G , where f is normalized and univalent and $f(G)$ is a circle domain. At each marked end, its distinguished target disks converge to the entire component $K_b(f)$.*
- (iv) *All complementary disks of Ω_n , and all images of physical points in Q_n° with $|z| \leq 4R$, lie in $|w| < 5R$. Any prescribed compact source sets, including finitely many smooth contours and their collars at each stage, can be retained in K_n . A retained oriented dust contour has the bounded-side correspondence of Lemma 3.3. For each fixed such contour its bounded target sides lie in a fixed disk, uniformly in n and in the number of retained tests.*

The finite subsets I_n carry their inherited order from $\mathcal{L}$; they need not be initial segments of its well-order. The original coordinates, their masks, and their budgets are unchanged.

Proof. First selection with the seed retained. Start the selection in Section 6 with $\mathcal{L}_0$ and Σ_0 . Every application of Theorem 6.3 permits precisely this arbitrary current countable list and permits a countable flag enlargement. At a successor stage append the selected barrier after the existing well-order. At a countable limit stage take the order-preserving union of all earlier lists and the union of the flags. These unions are countable. Each compatibility requirement concerns one new label and finitely many lower labels, so it already holds at a previous stage; increasing the flags can only remove free points. Thus compatibility persists, while all previously chosen functions and their bounds remain fixed.

The countable-basis argument in Section 6 makes this process stop at a countable ordinal. It produces a countable set S_1 as in Equation (6.4), with $K_b(f)$ a singleton for every $b \notin S_1$ in every later compatible transfer limit. Put $S = S_0 \cup S_1$. This enlargement is countable and contains the seed singularities. The newly selected barriers are real and introduce no singularities. In particular, Theorem 6.5 applies outside S .

Second selection at every mandatory mark. Consider the second-countable disjoint union

$$\mathcal{Q} = \coprod_{b \in S} \mathcal{Q}_b.$$

Starting with an empty excluded open subset of $\mathcal{Q}$, select an uncovered bad pair whenever one exists. Apply Lemma 7.4, append its coordinate, and add its excluded relative neighborhood to the open set. At countable limit stages take the order-preserving unions of coordinate lists and flags. All newly introduced period singularities belong to S .

This selection also stops at a countable stage. Each strict increase of the excluded open set contains a basic open set not contained in the preceding one. Once chosen, that basic set is contained in all later excluded open sets, so it cannot be chosen again. An ω_1 -long selection would contradict the countability of the basis. At termination every bad pair is covered, and the accumulated list and flags are countable. The exclusions persist by Lemma 7.4.

Finite transfers and preservation of the original tests. Enumerate the underlying countable set of final labels and take increasing finite subsets I_n exhausting it, each ordered by the inherited well-order. A countable well-order need not have finite initial segments exhausting it; none are needed here. Similarly choose increasing finite subsets exhausting S and include in the n th prescribed terminal indices every singularity of I_n . Apply Theorem 3.1 to I_n , with error ε_n , prescribed dust-cover mesh tending to zero, and input core containing H_n , the preceding core, and all additionally specified compact source sets. One may include compact neighborhoods of these sets, so the resulting cores are nested and contain H_n in their interiors. Include the fixed exterior region and all current mask boundaries as required by that theorem.

Finite circle-domain uniformization gives f_n and Ω_n . Exact core agreement, the total tail bound, and the individual positive-excess bounds are exactly the conclusions of Theorem 3.1. They apply to every fixed seed coordinate for all sufficiently large n . No selected flag enlargement changes a coordinate's original band budget. The terminal disks shrink in the quotient metric by the prescribed cover meshes. Normalized compactness and the uniform radius bound follow from Lemma 3.2.

Identification of the limiting components. Pass to a subsequence on which the maps and all marked disks converge. Lemma 3.3 gives $D_b \subset K_b(f)$ for $b \in S$. If one inclusion were strict, (f, D_b) would be a bad pair in a previously excluded neighborhood. The fixed coordinate responsible for that neighborhood occurs in every sufficiently late finite sublist. Its exact core values and vanishing transfer errors contradict its exclusion estimate. Thus $D_b = K_b(f)$ for every $b \in S$. All components outside S are points by Theorem 6.5, and these are all components by Lemma 3.3. Hence $f(G)$ is a circle domain.

Every argument just used depends on fixed retained coordinates and marks and survives further compatible extensions. A new marked end outside S already has a singleton component; its distinguished disk is contained in that singleton and therefore equals it. This also proves the stated persistence for additional singularities and marks.

Finally, retain a fixed contour and its collar in all sufficiently late cores. Orientation and Jordan separation give its bounded-side correspondence in each finite surface. For the bound on its image, choose $r_C > 2R$ so large that the source contour is inside $|z| < r_C$. The exterior estimate in Equation (3.16) bounds $f_n(\{|z| = r_C\})$ uniformly; the contour's image and its bounded side lie inside that outer image curve. This gives a bound depending only on R and the fixed contour, independent of the number of tests. Contours inside $|z| \leq 4R$ have the common bound $5R$. $\square$

Proof of Theorem 1.1. For a proper sphere domain make an initial Möbius change of source coordinate so that infinity belongs to the domain. Apply Theorem 7.5 with the empty seed list and no prescribed marks. It supplies a conformal circle-domain image. The whole-sphere case is immediate. $\square$

The following form uses continuous real tests on the quotient sphere. It is the retained-test input used in the rigidity companion. Its trace hypothesis is especially simple: every pair of boundary values has a planar image of area zero.

Corollary 7.6 (Retained scalar tests). *Let $G \subset \hat{\mathbb{C}}$ contain infinity, let $X = G \cup B$ be its quotient sphere, and fix R with $\{|z| \geq R\} \cup \{\infty\} \subset G$. Let $u_1, \dots, u_m \in C(X, \mathbb{R})$ be smooth on G , with $\int_G |du_i|^2 < \infty$, and suppose*

$$\text{area}((u_i, u_j)(B)) = 0 \quad (i \neq j). \quad (7.18)$$

For every positive sequence $\varepsilon_n \rightarrow 0$ there are finite bordered completions (Q_n, g_n) , nested smooth cores $K_n \subset G$ containing every compact subset of G in their interiors eventually, normalized conformal maps $f_n : Q_n^\circ \rightarrow \Omega_n$ onto finite circle domains, and smooth real tests $v_{i,n}$ on Ω_n such that

$$v_{i,n} \circ f_n = u_i \quad \text{near } K_n, \quad (7.19)$$

$$\sum_{i=1}^m \int_{\Omega_n \setminus f_n(K_n)} |\nabla v_{i,n}|^2 da < \varepsilon_n^2, \quad (7.20)$$

$$\sum_T A_{i,n}(T)^2 < \varepsilon_n^2 \quad (1 \leq i \leq m). \quad (7.21)$$

Here T ranges over the complementary round disks of Ω_n ; the cluster values of $v_{i,n}$ at T lie in an interval of length $A_{i,n}(T)$. The metric g_n is the original metric near K_n . After passage to a subsequence, $f_n \rightarrow f$ locally uniformly on G , where f is normalized and univalent and $f(G)$ is a circle domain. All finite target holes lie in $|w| < 5R$. The source-contour retention, correspondence, and bounds of Theorem 7.5(iv) also hold. In particular, these conclusions allow arbitrarily many, and possibly uncountably many, point components in the limiting complement.

Proof. If $B = \emptyset$, take $Q_n = K_n = G = \widehat{\mathbb{C}}$, f_n the identity, and $v_{i,n} = u_i$; all tail and oscillation sums vanish. We may therefore assume that B is nonempty.

Each test is bounded by compactness of X , has full mask, and has singleton band $\{u_i(b)\}$ at every $b \in B$. It is therefore a coordinate with budget $a_i = 0$, and the initial flag set may be empty. We verify compatibility in any chosen ordering of the finite list.

For distinct labels i, j , Equation (7.18) implies that the image of B under (u_i, u_j) modulo one has zero Haar measure on $\mathbb{T}^2$: decompose $\mathbb{R}^2$ into countably many integer translates of a unit square. For a label j and a finite tuple of lower labels $i_1, \dots, i_d$, a common dust point between $H_j(t)$ and any $H_{i_k}(t_k)$ requires (t, t_k) to lie in this null pair image. Each such condition is a null cylinder in $\mathbb{T}^{d+1}$, by Fubini. Their finite union is null, including when lower labels are repeated. Almost every tuple therefore has no such common dust point at all, which is stronger than compatibility (P).

Apply Theorem 7.5 with these seeds, choosing every finite sublist to contain all m seeds from the outset. Put $v_{i,n} = u_{i,n} \circ f_n^{-1}$. Exact core agreement gives Equation (7.19), and conformal invariance of energy gives Equation (7.20). Since all seed budgets vanish, Equation (7.17) gives Equation (7.21). The cluster-interval assertion is Lemma 3.3. All remaining conclusions are those of the theorem. $\square$

8 A hyperbolic convex-hull realization

We apply Luo and Wu's Theorem 1.1(a) [13] to the circle domain supplied by Theorem 1.1. Identify $\widehat{\mathbb{C}}$ with the ideal sphere $\partial_\infty \mathbb{H}^3$. A compact subset of that sphere is *circle type* if its complement is a circle domain. Write $C(Y)$ for the hyperbolic convex hull of such a set Y . For a three-dimensional hull, the intrinsic path distance on its boundary is the infimum of hyperbolic lengths of boundary curves joining the points.

Corollary 8.1 (Complete hyperbolic surfaces of genus zero). *Let S be a connected orientable surface without boundary, of genus zero, and let g be a complete Riemannian metric on S with Gaussian curvature identically -1 . There is a compact set $Y \subset \partial_\infty \mathbb{H}^3$, containing at least three points, whose complement is nonempty and connected, such that every connected component of Y is a closed round disk or a singleton and (S, d_g) is isometric to $\partial C(Y)$ with its intrinsic path metric.*

When $C(Y)$ is two-dimensional, $\partial C(Y)$ denotes the metric double of $C(Y)$ across its relative boundary in the containing totally geodesic plane. Surfaces are understood to be Hausdorff and second countable; their number of ends is unrestricted.

Proof. The orientation and metric make S a genus-zero Riemann surface. Its classical planar realization, recalled in [13, Section 1.1], and Theorem 1.1 give a conformal equivalence from S to a circle domain V . The transported metric is complete with curvature -1 , so it is the unique complete conformal hyperbolic metric of V . In particular, V is hyperbolic and its boundary has at least three points. Luo and Wu's Theorem 1.1(a), with their two-dimensional convention in

Section 1.1, now gives the asserted intrinsic isometry. The realizing set has at least three points, since the convex hull of at most two ideal points cannot supply a hyperbolic surface. $\square$

The realizing set Y need not be the complement of the intermediate circle domain V . The corollary asserts existence and intrinsic isometry; it supplies neither uniqueness nor a prescribed extension to the ideal boundary. Its curvature hypothesis is exactly -1 . The realization of arbitrary complete metrics of curvature at least -1 with each end represented by a circle or point at infinity, posed in [13, Section 1.2, Conjecture 3], is a separate problem.

References

- [1] Lars V. Ahlfors, *Conformal Invariants: Topics in Geometric Function Theory*, McGraw-Hill Series in Higher Mathematics, McGraw-Hill, New York, 1973. Reprinted by the American Mathematical Society, 2010. [Publisher record](#).
- [2] Luigi Ambrosio, Simone Di Marino, and Giuseppe Savaré, On the duality between p -modulus and probability measures, *J. Eur. Math. Soc.* **17** (2015), no. 8, 1817–1853. [doi:10.4171/JEMS/546](#).
- [3] John B. Conway, *Functions of One Complex Variable II*, Graduate Texts in Mathematics, vol. 159, Springer-Verlag, New York, 1995. [doi:10.1007/978-1-4612-0817-4](#).
- [4] Behnam Esmayli and Kai Rajala, Conformal uniformization of domains bounded by quasitripods, *Duke Math. J.* **175** (2026), no. 4, 555–605. [doi:10.1215/00127094-2025-0035](#). [Preprint arXiv:2401.08485v2](#).
- [5] Herbert Federer, *Geometric Measure Theory*, Grundlehren der mathematischen Wissenschaften, vol. 153, Springer-Verlag, 1969. Reprinted in *Classics in Mathematics*, Springer-Verlag, 1996. [doi:10.1007/978-3-642-62010-2](#).
- [6] Bent Fuglede, Extremal length and functional completion, *Acta Math.* **98** (1957), 171–219. [doi:10.1007/BF02404474](#).
- [7] Zheng-Xu He and Oded Schramm, Fixed points, Koebe uniformization and circle packings, *Ann. of Math. (2)* **137** (1993), no. 2, 369–406. [doi:10.2307/2946541](#).
- [8] Juha Heinonen and Pekka Koskela, Quasiconformal maps in metric spaces with controlled geometry, *Acta Math.* **181** (1998), 1–61. [doi:10.1007/BF02392747](#).
- [9] Christina Karafyllia and Dimitrios Ntalampekos, Uniformization of Gromov hyperbolic domains by circle domains, *Duke Math. J.* **175** (2026), no. 13, 2495–2530. [doi:10.1215/00127094-2025-0076](#). [Preprint arXiv:2405.13782](#), 2024.
- [10] Alexander S. Kechris, *Classical Descriptive Set Theory*, Graduate Texts in Mathematics, vol. 156, Springer-Verlag, New York, 1995. [doi:10.1007/978-1-4612-4190-4](#).
- [11] Paul Koebe, Über die Uniformisierung beliebiger analytischer Kurven. (Dritte Mitteilung), *Nachr. Ges. Wiss. Göttingen, Math.-Phys. Kl.* (1908), 337–358. [Digitized article record](#).
- [12] Paul Koebe, Abhandlungen zur Theorie der konformen Abbildung. VI. Abbildung mehrfach zusammenhängender schlichter Bereiche auf Kreisbereiche. Uniformisierung hyperelliptischer Kurven. (Iterationsmethoden), *Math. Z.* **7** (1920), 235–301. [doi:10.1007/BF01199400](#). [Digitized article record](#).
- [13] Feng Luo and Tianqi Wu, The Koebe conjecture and the Weyl problem for convex surfaces in hyperbolic 3-space, *Adv. Math.* **458** (2024), Part B, Paper No. 109969. [doi:10.1016/j.aim.2024.109969](#). [Preprint arXiv:1910.08001](#).
- [14] R. L. Moore, Concerning upper semi-continuous collections of continua, *Trans. Amer. Math. Soc.* **27** (1925), no. 4, 416–428. [doi:10.1090/S0002-9947-1925-1501320-8](#).

- [15] Dimitrios Ntalampekos, *Uniformization problems in the plane: A survey*, preprint, 2026. [arXiv:2603.15098](#).
- [16] Dimitrios Ntalampekos and Kai Rajala, Exhaustions of circle domains, *Int. Math. Res. Not. IMRN* (2025), no. 19, Paper No. rnaf296, 25 pp. [doi:10.1093/imrn/rnaf296](#). Preprint [arXiv:2312.06840](#).
- [17] OpenAI, Removable Boundaries and Rigidity of Circle Domains, OpenAI Math Release preprint [OAI:Removable-Boundaries-and-Rigidity-of-Circle-Domains-September-23-2026](#), 2026.
- [18] Kai Rajala, Uniformization of planar domains by exhaustion, *Amer. J. Math.* **147** (2025), no. 4, 1025–1051. [doi:10.1353/ajm.2025.a966291](#).
- [19] Oded Schramm, Transboundary extremal length, *J. Analyse Math.* **66** (1995), 307–329. [doi:10.1007/BF02788827](#).
- [20] Maurice Sion, On general minimax theorems, *Pacific J. Math.* **8** (1958), no. 1, 171–176. [Publisher article](#).
- [21] G. T. Whyburn, *Analytic Topology*, American Mathematical Society Colloquium Publications, vol. 28, American Mathematical Society, 1942. [Publisher record](#).